
SC-FAGC: Size Constrained Fast Anchor Graph Clustering

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Abstract

Spectral clustering is hindered by its $O(n^3)$ time complexity, and although anchor-based approximations reduce this to $O(nm^2)$ ($m \ll n$), they often yield degenerate or unbalanced partitions in the absence of explicit cluster-size control. To address this limitation, we propose **SC-FAGC (Size-Constrained Fast Anchor Graph Clustering)**, a unified formulation that integrates entropy regularization and bilateral cardinality constraints within an anchor-based spectral clustering framework, enforcing user-specified lower and upper bounds on anchor-level cluster sizes to avoid trivial or highly unbalanced solutions. To solve the resulting non-convex problem efficiently, we develop an **Iteratively Re-weighted (IRW)** optimization scheme that linearizes the objective and solves each subproblem via a **Double-Bounded Optimal Transport (DB-OT)** solver based on the **Sinkhorn–Knopp** algorithm, guaranteeing convergence while maintaining scalability. Extensive experiments on benchmark datasets demonstrate that SC-FAGC consistently achieves state-of-the-art performance across six standard clustering metrics while strictly enforcing the prescribed anchor-level constraints. Moreover, it induces effective data-level size control after label transmission, offering a principled and scalable solution for large-scale graph clustering with controllable partition structure.

1. Introduction

Clustering is a fundamental technique in data mining and machine learning, aiming to partition data points into distinct groups such that points within the same group are similar while those in different groups are dissimilar (MacQueen, 1967). Among various clustering approaches, Spectral Clustering (SC) (Shi & Malik, 2000; Luxburg, 2007; Ding et al.,

2024) has garnered significant attention due to its ability to capture complex non-linear structures and its strong theoretical foundation in graph partitioning. However, standard SC involves constructing an $n \times n$ affinity matrix and performing eigendecomposition, leading to a computational complexity of $O(n^3)$, where n is the number of data points. This cubic complexity severely hinders the application of SC to large-scale (Chen et al., 2020) datasets.

To address this scalability bottleneck, anchor-based spectral clustering methods have been extensively developed (Zhang et al., 2018; Wang et al., 2017). These methods construct a bipartite graph (Wang et al., 2024a) between the original data points and a small set of m representative anchors ($m \ll n$), thereby reducing the computational complexity from $O(n^3)$ to $O(nm^2)$. Representative algorithms include Landmark-based Spectral Clustering (LSC) (Chen & Cai, 2011) and Fast Spectral Clustering (FSC) (Zhu et al., 2017). Subsequent works have further improved the efficiency and robustness of anchor graph construction, utilized effectively in methods like Ultra-Scalable Spectral Clustering (USPEC) (Huang et al., 2020) and Fast Rank-One Weighting Learning (FRWL) (Wang et al., 2020). More recently, enhanced anchor-based strategies such as EDCAG (Wang et al., 2024c), FCAG (Nie et al., 2024c), LFCAG (Wang et al., 2024b), and anchor-guided label spreading for co-clustering (Xie et al., 2025a) have been proposed to better capture the underlying data manifold. Complementary to anchor-based acceleration, nearly-linear spectral clustering has also advanced for unconstrained graph settings: Peng et al. (Peng et al., 2015) established guarantees for graphs satisfying well-clustering conditions, and Fast and Simple Spectral Clustering (FSSC) (Macgregor, 2023) replaces the full k -eigenvector embedding with an $O(\log k)$ power-method embedding, achieving nearly-linear runtime while maintaining competitive clustering quality. Deep learning approaches such as SpectralNet (Shaham et al., 2018) and self-supervised methods like SCGC (Kulatilleke et al., 2024) have also emerged for learning spectral embeddings end-to-end.

Despite their success in improving efficiency, most existing anchor-based clustering methods suffer from a critical limitation: they often lack explicit control over the size of the resulting clusters. In many real-world applications, it is desirable or even necessary to avoid trivial

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partitions where some clusters are excessively large while others contain very few points. While some studies have explored balanced clustering, such as Balanced K-Means with Novel Constraint (BKNC) (Chen et al., 2022) and Fast Clustering with Flexible Balance Constraints (FCBC) (Liu et al., 2018), as well as scalable methods with group fairness constraints (Wang et al., 2023), these are often limited to specific objective functions or lack a unified spectral formulation. In multi-view scenarios, anchor learning with cluster constraints has also been investigated (Chen et al., 2025). Earlier optimization-based graph partitioning work addressed cardinality constraints through continuous quadratic programming (CQP) formulations (Hager & Krylyuk, 1999); this line is particularly relevant because it is the closest classical cardinality-aware baseline to our setting, although it predates modern anchor-graph spectral pipelines. Recent attempts to incorporate size constraints into anchor graph clustering, such as FAGC-R (Xie et al., 2025b), EAGC (Xie et al., 2025d), and other min-cut variations (Nie et al., 2024b), have shown promise but often face challenges in efficiently solving the resulting constrained optimization problems.

In this paper, we propose a novel framework termed **SC-FAGC (Size-Constrained Fast Anchor Graph Clustering)** to bridge this gap. SC-FAGC integrates entropy regularization (Peyré & Cuturi, 2019) and explicit bilateral cardinality constraints into the anchor-based spectral clustering objective. This formulation not only promotes the smoothness of the assignment matrix but also ensures that each anchor-level cluster mass falls within a user-specified range, effectively preventing trivial solutions. To solve the resulting non-convex optimization problem, we employ an **Iteratively Re-weighted (IRW)** algorithm, which transforms the problem into a sequence of convex subproblems. Each subproblem is then efficiently solved using a **Double-Bounded Optimal Transport (DB-OT)** approach (Shi et al., 2024), leveraging the Generalized Sinkhorn-Knopp algorithm (Cuturi, 2013). Extensive experiments on various benchmark datasets demonstrate that SC-FAGC achieves state-of-the-art clustering performance while enforcing anchor-level size constraints and inducing practical data-level size control through label transmission.

Conflict of Interest Disclosure. The authors declare no financial or non-financial conflicts of interest related to this work. All experiments are conducted on publicly available benchmark datasets, and no proprietary systems or models developed by the authors’ affiliations are evaluated.

2. Proposed Method

In this section, we present the proposed **SC-FAGC** framework. We first review the preliminaries of spectral cluster-

ing and anchor graph construction, then detail our objective function with size constraints, and finally describe the efficient optimization algorithm based on optimal transport.

Nomenclature

We summarize the key notations used throughout this paper in Table 1.

Table 1. Summary of Notations.

SYMBOL	DESCRIPTION
$G = (V, E)$	Undirected weighted graph, $ V = n$
$W \in \mathbb{R}^{n \times n}$	Symmetric non-negative adjacency matrix
$L \in \mathbb{R}^{n \times n}$	Graph Laplacian matrix
m	Number of anchor points ($m \ll n$)
$Z \in \mathbb{R}^{m \times d}$	Feature matrix of anchors
$B \in \mathbb{R}^{n \times m}$	Bipartite affinity matrix (Points $\leftrightarrow$ Anchors)
$A \in \mathbb{R}^{m \times m}$	Anchor-level similarity matrix, $A = B^\top B$
c	Number of clusters
$U \in \mathbb{R}^{m \times c}$	Soft cluster assignment matrix for anchors
α, β	Lower and upper bounds on cluster sizes
ϵ	Entropy regularization parameter
$H(U)$	Entropy regularizer: $-\sum_{i,j} u_{ij} \log(u_{ij})$
$P \in \mathbb{R}^{m \times c}$	Gibbs kernel: $P = \exp(D/\epsilon)$, where $D = 2AU$

2.1. Standard Spectral Clustering and Its Limitations

Consider an undirected weighted graph $G = (V, E)$ with $|V| = n$. Let $W \in \mathbb{R}^{n \times n}$ be a symmetric nonnegative adjacency matrix and define the degree matrix as $S = \text{diag}(W\mathbf{1})$. The graph Laplacian matrix L takes different forms depending on the cut criterion:

- For **RatioCut**: $L = S - W$.
- For **NormalizedCut**: $L = S^{-1/2}(S - W)S^{-1/2}$.

Spectral clustering can be viewed as a spectral relaxation of the graph partitioning problem:

$$\min_{F^\top M F = I} \text{Tr}(F^\top L F), \quad (1)$$

where $M = I$ for RatioCut and $M = S$ for Normalized Cut. The solution is obtained by computing the k eigenvectors corresponding to the smallest k eigenvalues of L , forming the embedding matrix $F \in \mathbb{R}^{n \times k}$.

However, this approach suffers from high computational complexity: the eigendecomposition requires $O(kn^2) \sim O(n^3)$ time and $O(n^2)$ memory, which becomes prohibitive for large-scale graphs.

2.2. Anchor-Based Bipartite Graph Construction

To alleviate this computational bottleneck, we select m anchor points ($m \ll n$) represented by the matrix $Z \in$

$\mathbb{R}^{m \times d}$, and construct a bipartite affinity matrix $B \in \mathbb{R}^{n \times m}$ using a parameter-free neighbor assignment strategy. For each sample x_i , its soft assignment to anchors is obtained by solving:

$$\min_{\substack{b^i \mathbf{1}_m = \mathbf{1} \\ b^i \geq 0}} \sum_{j=1}^m \|x_i - z_j\|_2^2 b_{ij} + \gamma \sum_{j=1}^m b_{ij}^2, \quad (2)$$

where $\gamma > 0$ is a regularization parameter. This problem admits a closed-form solution: for each x_i , let $\{z_{(1)}, \dots, z_{(k)}\}$ be the k -nearest anchors sorted by increasing distance, and define $d_{ij} = \|x_i - z_j\|_2^2$. Then,

$$b_{ij} = \begin{cases} \frac{d_{i,(k+1)} - d_{ij}}{kd_{i,(k+1)} - \sum_{h=1}^k d_{i,(h)}}, & j \leq k, \\ 0, & j > k. \end{cases} \quad (3)$$

The Lagrangian and KKT derivation of Eq. (3), including the elimination of γ under the simplex constraint, is provided in Appendix K.

By construction, $B\mathbf{1}_m = \mathbf{1}_n$, so B can be interpreted as a probability matrix. We then perform a one-mode projection to obtain the anchor-level similarity matrix:

$$A = B^\top B. \quad (4)$$

Since $(A)_{j_1 j_2} = \sum_{i=1}^n b_{ij_1} b_{ij_2}$, the matrix A captures the co-assignment relationships among anchors. Consequently, the clustering task is reduced from n samples to m anchors, reducing complexity from $O(n^3)$ to $O(nm^2)$.

2.3. Objective Function

To encourage high intra-cluster similarity among anchors while preventing trivial solutions, we propose **SC-FAGC**, which explicitly incorporates bilateral size constraints and entropy regularization. We seek a soft assignment matrix $U \in \mathbb{R}^{m \times c}$ for the anchors, where u_{ij} represents the probability of anchor i belonging to cluster j .

The core objective is to maximize the intra-cluster compactness on the anchor graph while enforcing size bounds and assignment smoothness. Our optimization problem is formulated as:

$$\begin{aligned} \max_U \quad & \text{Tr}(U^\top (B^\top B) U) + \epsilon H(U) \\ \text{s.t.} \quad & U\mathbf{1}_c = \mathbf{1}_m, \quad U \geq 0, \\ & \alpha \cdot \mathbf{1}_c \leq U^\top \mathbf{1}_m \leq \beta \cdot \mathbf{1}_c, \end{aligned} \quad (5)$$

where:

- $\text{Tr}(U^\top (B^\top B) U)$ measures the spectral connection strength within clusters.

- $H(U) = -\sum_{i,j} u_{ij} \log u_{ij}$ is the entropy regularization term with weight $\epsilon > 0$, avoiding hard assignment instability and broadening the optimization landscape.
- The constraints $U\mathbf{1}_c = \mathbf{1}_m$ and $U \geq 0$ ensure U is a valid row-stochastic probability matrix.
- The bilateral constraints $\alpha \leq \sum_{i=1}^m u_{ij} \leq \beta$ ensure that each anchor-level cluster mass lies within the range $[\alpha, \beta]$, preventing cluster collapse during anchor-level optimization.

The feasibility guarantee is exact for U . After label transmission, data-level size control is induced through the bipartite graph and is therefore an approximate propagated property rather than a hard feasibility constraint on individual samples.

2.4. Optimization Strategy

The objective in Eq. (5) is non-convex due to the quadratic trace term maximizing a convex function. We employ an **Iteratively Re-weighted (IRW)** scheme to handle this.

Step 1: Linearization. At iteration t , we linearize the quadratic term $\text{Tr}(U^\top A U)$ at the current point U . The gradient is $D = 2AU$. Maximize the linearized surrogate is equivalent to maximizing $\text{Tr}(D^\top U)$.

Step 2: Solving the Subproblem. The subproblem at iteration t becomes:

$$\max_{U \in \mathcal{C}} \langle D, U \rangle + \epsilon H(U), \quad (6)$$

where $\mathcal{C}$ encapsulates the row-sum and bilateral column-sum constraints. By negating the objective, this is equivalent to an Entropic Optimal Transport problem with inequality constraints on the marginals, known as **Double-Bounded Optimal Transport (DB-OT)** (Shi et al., 2024).

The solution is given by the scaling form $U^* = \text{diag}(\mathbf{u}) P \text{diag}(\mathbf{q} \odot \mathbf{v})$, where $P = \exp(D/\epsilon)$ is the Gibbs kernel. The scaling vectors $\mathbf{u}, \mathbf{q}, \mathbf{v}$ can be found efficiently via a generalized **Sinkhorn-Knopp** (Cuturi, 2013) algorithm.

2.4.1. THEORETICAL ANALYSIS

In this section, we provide a rigorous theoretical analysis of the proposed framework. We explicitly characterize the convergence properties of the Iteratively Re-weighted (IRW) scheme and the optimality conditions of the solution. We show that our algorithm falls within the Minorization-Maximization (MM) framework, guaranteeing convergence (Lacoste-Julien, 2016) to a stationary point of the non-convex objective function.

Problem Setup and Geometry. Let $\mathcal{C} \subset \mathbb{R}^{m \times c}$ denote the feasible set defined by the intersection of the probability simplex and the bilateral constraints:

$$\mathcal{C} = \{U \in \mathbb{R}^{m \times c} \mid U \geq 0, U\mathbf{1}_c = \mathbf{1}_m, \alpha\mathbf{1}_c \leq U^\top \mathbf{1}_m \leq \beta\mathbf{1}_c\}. \quad (7)$$

The set $\mathcal{C}$ is a compact convex polytope. The objective function defined in Eq. (5) can be written as:

$$\mathcal{J}(U) = f(U) + \epsilon H(U), \quad (8)$$

where $f(U) = \text{Tr}(U^\top AU)$ represents the cohesion term, and $H(U)$ is the entropy. Since $A = B^\top B \succeq 0$, $f(U)$ is a convex function. Consequently, maximizing $\mathcal{J}(U)$ involves maximizing a convex function over a convex set, which is known to be non-convex and generally NP-hard to find the global maximum. However, we prove that our iterative scheme guarantees convergence to a valid stationary point.

Lemma 2.1 (Surrogate Function Construction). *Let $U^{(t)}$ be the estimate at iteration t . Consider the surrogate function:*

$$\mathcal{G}(U \mid U^{(t)}) = \text{Tr}(2U^{(t)\top} AU) - \text{Tr}(U^{(t)\top} AU^{(t)}) + \epsilon H(U). \quad (9)$$

Then, $\mathcal{G}(U \mid U^{(t)})$ is a valid minorizer of $\mathcal{J}(U)$ at $U^{(t)}$, satisfying: (i) $\mathcal{G}(U^{(t)} \mid U^{(t)}) = \mathcal{J}(U^{(t)})$; (ii) $\mathcal{G}(U \mid U^{(t)}) \leq \mathcal{J}(U)$ for all $U \in \mathcal{C}$.

The proof is provided in Appendix L.

Based on Lemma 2.1, the update rule in our algorithm corresponds to maximizing the minorizer:

$$\begin{aligned} U^{(t+1)} &= \arg \max_{U \in \mathcal{C}} \mathcal{G}(U \mid U^{(t)}) \\ &= \arg \max_{U \in \mathcal{C}} \langle D^{(t)}, U \rangle + \epsilon H(U), \end{aligned} \quad (10)$$

where constants independent of U are omitted, and $D^{(t)} = 2AU^{(t)}$.

Theorem 2.2 (Monotonic Ascent and Convergence). *The sequence of objective values $\{\mathcal{J}(U^{(t)})\}_{t \geq 0}$ generated by Algorithm 1 is strictly non-decreasing (unless stationary) and converges.*

The proof is provided in Appendix L.

Theorem 2.3 (First-Order Optimality). *Any limit point U^* of the sequence $\{U^{(t)}\}$ generated by the algorithm is a stationary point of the original problem (5) satisfying the first-order KKT (Dutta et al., 2013) conditions.*

The proof is provided in Appendix L.

Proposition 2.4 (Uniqueness and Sinkhorn Convergence). *The linearized subproblem (6) is strictly concave due to the*

Algorithm 1 Optimization Algorithm for SC-FAGC

- 1: **Input:** Bipartite graph B , clusters c , bounds α, β , parameter ϵ .
 - 2: Compute anchor similarity $A = B^\top B$.
 - 3: Initialize $U \in \mathbb{R}^{m \times c}$ by N^2HI .
 - 4: // N^2HI initializes U on $B^\top B$; anchors are generated separately by $BKHK$.
 - 5: **repeat**
 - 6: Compute gradient $D = 2AU$.
 - 7: Define kernel $P = \exp(D/\epsilon)$.
 - 8: // Sinkhorn Loop for Subproblem
 - 9: Initialize $\mathbf{u} \leftarrow \mathbf{1}_m, \mathbf{v} \leftarrow \mathbf{1}_c, \mathbf{q} \leftarrow \mathbf{1}_c$.
 - 10: **repeat**
 - 11: Update $\mathbf{u} \leftarrow \mathbf{1}_m \odot (P(\mathbf{v} \odot \mathbf{q}))$.
 - 12: Update $\mathbf{q} \leftarrow \max(\alpha\mathbf{1}_c \odot ((P^\top \mathbf{u}) \odot \mathbf{v}), \mathbf{1}_c)$.
 - 13: Update $\mathbf{v} \leftarrow \min(\beta\mathbf{1}_c \odot ((P^\top \mathbf{u}) \odot \mathbf{q}), \mathbf{1}_c)$.
 - 14: **until** Sinkhorn convergence
 - 15: Update $U^{(t+1)} = \text{diag}(\mathbf{u})P\text{diag}(\mathbf{q} \odot \mathbf{v})$.
 - 16: **until** Global convergence
 - 17: **Output:** Final assignments U^* .
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entropic regularization term $\epsilon H(U)$. Provided the feasible set $\mathcal{C}$ is non-empty, there exists a unique global maximizer $U^{(t+1)}$. The Sinkhorn-Knopp iteration converges to this unique solution at a linear rate.

The proof is provided in Appendix L.

The complete algorithm is summarized in Algorithm 1.

Table 2. Datasets Description

Type	Datasets	Samples	Dimension	Cluster
Toy Data	Aggregation	788	2	7
	Flame	240	2	2
	Two ring	500	2	2
	Two moon	400	2	2
Real-world Data	Jaffe	213	1024	10
	Ar	1400	2200	100
	Mpeg7	1400	6000	70
	Msra25	1799	256	12
	Usps20	1854	256	10
	Palm25	2000	256	100
	Segment	2310	19	7
	Waveform21	2746	21	3
	Mnist10	6996	784	10
	Coil100	7200	1024	100
	Isolet	7797	617	26
	Letter	20000	16	26

3. Experiments

We conduct extensive experiments on both synthetic and real-world datasets to evaluate the effectiveness of the proposed SC-FAGC. Specifically, we first employ four synthetic datasets to intuitively visualize the clustering performance on various manifold structures. Then, we perform com-

prehensive comparisons on twelve real-world benchmarks against state-of-the-art methods. We further analyze the sensitivity of the model to key hyperparameters, including the number of anchors and neighbors. The convergence behavior of the optimization algorithm is also demonstrated. Moreover, we verify the effectiveness of the bilateral constraints by examining the resulting cluster size distributions. Finally, the execution time is reported to evaluate the computational efficiency.

3.1. Toy Example

In the toy experiment, four synthetic datasets, namely Aggregation, Two Moon, Flame, and Two Ring, are utilized. These datasets pose unique challenges for clustering algorithms due to their distinct shapes and manifolds. The detailed information of these synthetic datasets is displayed in Table 2. We compare our proposed method SC-FAGC with four commonly used algorithms: KM, FCM, NCut, and LSC. The visualization of clustering results is presented in Fig. 1.

For Aggregation, KM and FCM perform poorly as they tend to partition data into spherical shapes, failing to capture the irregular connections. NCut and LSC also exhibit suboptimal performance. In contrast, SC-FAGC achieves nearly perfect clustering by effectively capturing the global structure while adhering to size constraints. For the Two Moon dataset, KM and FCM cluster the data with a hyperplane, failing to capture the nonlinear manifold structure. NCut and LSC perform significantly better, but SC-FAGC achieves a perfect accuracy of 1.000. On the Flame dataset, KM and FCM struggle with the non-Gaussian stratified structure. LSC performs well, but SC-FAGC further refines the result to 0.996. For the Two Ring dataset, which contains concentric circles, KM and FCM fail completely as they cannot handle non-convex clusters. LSC also struggles (0.7512) likely due to anchor representation limits on such geometry. SC-FAGC, however, achieves perfect separation. All in all, our proposed SC-FAGC consistently captures the inherent structures of different types of data, yielding superior clustering results compared to baselines.

3.2. Real-world Datasets and Settings

In addition to visualization on synthetic datasets, we conduct experiments on 12 real-world datasets to further validate the effectiveness of the proposed method. The specific details regarding these datasets are presented in Table 2, encompassing face, digit, object, and biological data. Among them, the Segment, Waveform21, Isolet, and Letter datasets are obtained from the UCI Machine Learning Repository. The Mpeg7, MSRA25, Palm25, and COIL100 datasets are image data. Before clustering, to mitigate the influence of varying feature scales, all data features underwent normal-

ization to achieve a mean of zero and a standard deviation of one by utilizing z-score normalization.

Fourteen widely used clustering methods are compared to evaluate the performance of the proposed approach, including KM (MacQueen, 1967), NCut (Shi & Malik, 2000), BKNC (Chen et al., 2022), LSC (Chen & Cai, 2011), FRWL (Wang et al., 2020), FSC (Zhu et al., 2017), FCBC (Liu et al., 2018), USPEC (Huang et al., 2020), ED-CAG (Wang et al., 2024c), LFCAG (Wang et al., 2024b), FCAG (Nie et al., 2024c), FAGC-R (Xie et al., 2025b), CQP (Hager & Krylyuk, 1999), and Fast and Simple Spectral Clustering (FSSC) (Macgregor, 2023). KM is the typical partition-based clustering method, CQP is a cardinality-aware graph partitioning baseline, FSSC is a modern unconstrained nearly-linear spectral clustering baseline, and the other methods are graph-based clustering approaches, many of them anchor-based. The similarity graph in NCut is a k-nn graph, constructed using a Gaussian kernel function with a bandwidth equal to the average Euclidean distance of the data points. The range of values for the number of nearest neighbors is $\{5, 10, 15, 20, 25, 30, 35, 40\}$. For anchor-based methods, the number of anchors is set to 512 for the Mpeg7, MSRA25, and Palm25 datasets, while it is 1024 for all other datasets. In our implementation, the m anchors are generated by Balanced K-means based Hierarchical K-means (BKHK), which recursively bisects the data and is designed to encourage comparatively balanced anchor regions. The bipartite affinity matrix $B \in \mathbb{R}^{n \times m}$ is then constructed by the parameter-free k -nearest-anchor assignment in Eq. (3), so that $B\mathbf{1}_m = \mathbf{1}_n$, and the anchor similarity is computed as $A = B^\top B$. For subgraph construction, we set the number of neighbors to 5 for LSC and USPEC as suggested in their original papers, while for other anchor-based methods including ours, we search from $\{5, 10, 15, 20, 25, 30, 35, 40\}$. For the LFCAG method, the parameter λ ranges from 2 to 20 with an interval of 2. Under each parameter configuration, each algorithm is run ten times and the average results are recorded. To ensure stable and reproducible clustering performance, we adopt the Nearest Neighbor Hierarchical Initialization (N^2HI) method (Nie et al., 2024a) for initializing the discrete cluster assignments. Specifically, the initialization graph for our anchor-based method is set as $B^\top B$, which enables deterministic results and eliminates the variability typically associated with random initialization. We emphasize that N^2HI initializes the cluster assignment U , while BKHK is used for anchor generation.

Six clustering metrics are used to evaluate the performance of clustering algorithms: Accuracy (ACC), Normalized Mutual Information (NMI), Purity, Precision, Recall, and Adjusted Rand Index (ARI). ACC measures the degree of agreement between the predicted cluster labels and the ground truth labels after resolving label permutation ambiguity by

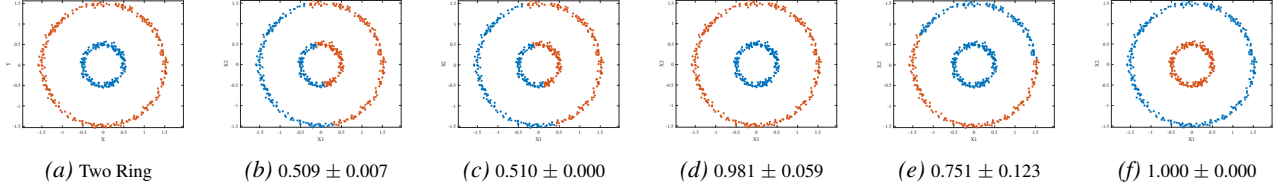


Figure 1. Visualization of clustering results on the Two Ring dataset. The first column shows the Ground Truth. The subsequent columns show results of KM, FCM, NCut, LSC, and SC-FAGC. Additional results on Aggregation, Two Moon, and Flame datasets are provided in Fig. 5 in the Appendix.

the Hungarian algorithm (linear sum assignment). The range of ACC is between 0 and 1, where a larger value indicates better clustering performance. NMI compares the similarity between the cluster assignments and the ground truth while accounting for chance, normalized by the entropy of the labels. Purity evaluates the homogeneity of clusters by measuring the proportion of data points in a cluster that belong to the majority class. Precision and Recall allow for a more granular view of the clustering quality in terms of pair-wise correctness. ARI is an adjusted measure of the Rand Index that corrects for chance, providing a robust measure of similarity between two data clusterings. For all metrics, higher values indicate better performance.

3.3. Clustering Performance

In this subsection, we compare the mean and standard deviation of clustering performance by different methods across multiple datasets. The clustering results are presented in Tables 3 through 8. Specifically, Table 3 corresponds to Accuracy (ACC), Table 4 to Normalized Mutual Information (NMI), Table 5 to Purity, Table 6 to Precision, Table 7 to Recall, and Table 8 to Adjusted Rand Index (ARI). The best-performing method in the tables is marked in red, while the second-best and third-best performing methods are indicated with orange and yellow, respectively. From the clustering results, the following observations can be made.

Firstly, NCut generally outperforms KM in clustering performance on most datasets. This suggests that graph-based clustering methods are better able to capture the intrinsic structural information of the data in most cases.

Secondly, anchor-based methods like LSC and USPEC often achieve competitive or superior performance compared to standard spectral clustering (NCut) and other baselines, demonstrating the effectiveness of the anchor graph representation.

Thirdly, the proposed SC-FAGC method consistently yields the best results on the majority of datasets. For instance, SC-FAGC achieves the highest clustering accuracy on the MSRA25, USPS20, Palm25, Mnist10, COIL100, Mpeg7, Isolet, Waveform21, and Jaffe datasets. On the USPS20 dataset, SC-FAGC reaches an accuracy of 88.83%, signifi-

cantly surpassing the second-best method.

Among the newly included strongly-related baselines, CQP is the closest to our target setting because it also performs cardinality-aware graph partitioning. As shown in Tables 3–8, SC-FAGC outperforms CQP on the majority of datasets (e.g., 10/12 in ACC and 11/12 in NMI), which we attribute to the soft anchor graph representation and the more flexible bilateral size bounds optimized on the reduced anchor graph. The main exceptions arise on datasets with near-uniform class sizes, where stricter equal-cardinality partitioning remains competitive.

FSSC provides a complementary comparison as a modern nearly-linear unconstrained spectral clustering method. SC-FAGC achieves consistently stronger clustering quality across the reported metrics; while FSSC is typically faster, as expected for an unconstrained nearly-linear design, this reflects a trade-off between efficiency and size-aware quality rather than a like-for-like replacement.

Finally, the stability of SC-FAGC is superior.

3.4. Sensitivity Analysis

In our SC-FAGC model, the entropy regularization parameter ϵ is fixed to 0.01 across all experiments, which provides sufficient smoothing for the optimal transport solver while maintaining clustering precision. In the construction of the bipartite graph, there are two parameters: the number of nearest neighbors and anchors. In this subsection, we explore the impact of these two factors on the clustering accuracy. The values of nearest neighbors are selected from $\{5, 10, 15, 20, 25, 30, 35, 40\}$. The range of number of anchors is selected from $\{2^7, 2^8, 2^9, 2^{10}, 2^{11}, 2^{12}\}$. Under each set of corresponding parameters, the algorithms are conducted ten times to avoid randomness and the average results are recorded. We visualize the sensitivity results as three-dimensional bar charts shown in Fig. 2. From Fig. 2, it can be observed that the clustering accuracy varies with both the number of anchor points and the number of neighbors. When the number of anchor points is small, the clustering accuracy decreases with an increase in the number of neighbors. For instance, when the quantity of anchor points is 2^7 , the clustering accuracy decreases with an increase

Table 3. Clustering Accuracy (ACC; Mean $\pm$ Std(%)) of Compared Methods on Real-World Datasets

Method	MSRA25	AR	Mnist10	USPS20	Palm25	COIL100	Mpeg7	Letter	Segment	Isolet	Waveform21	Jaffe
KM	49.12 _{4.30}	26.51 _{1.11}	53.98 _{1.92}	63.88 _{3.70}	68.39 _{3.26}	45.00 _{1.90}	47.30 _{1.91}	25.90 _{0.87}	61.88 _{4.21}	53.42 _{3.36}	50.72 _{0.02}	77.36 _{8.72}
NCut	54.93 _{3.70}	34.41 _{1.54}	69.96 _{5.16}	71.72 _{6.42}	74.11 _{2.28}	54.11 _{2.37}	51.73 _{1.19}	33.56 _{1.29}	56.18 _{4.97}	54.09 _{2.68}	52.37 _{0.00}	85.48 _{9.03}
BKNC	55.72 _{1.11}	31.62 _{0.79}	51.93 _{1.10}	68.03 _{1.66}	84.98 _{0.63}	50.95 _{1.06}	54.69 _{0.99}	27.95 _{1.83}	57.06 _{1.60}	50.96 _{0.94}	57.21 _{6.39}	94.46 _{1.02}
LSC	58.14 _{2.39}	33.92 _{0.81}	71.60 _{4.71}	74.18 _{4.63}	76.48 _{1.37}	50.50 _{0.76}	54.85 _{0.98}	33.18 _{1.12}	67.40 _{2.74}	57.95 _{1.81}	52.95 _{3.27}	93.11 _{4.82}
FRWL	61.00 _{2.53}	32.76 _{0.48}	66.84 _{2.39}	75.24 _{3.19}	77.92 _{0.28}	54.01 _{1.71}	52.57 _{1.30}	32.23 _{1.48}	62.12 _{0.41}	58.66 _{2.77}	59.43 _{0.00}	95.93 _{0.27}
FSC	59.31 _{3.68}	32.01 _{0.66}	65.10 _{5.05}	75.77 _{0.92}	75.78 _{1.23}	54.39 _{0.79}	53.19 _{2.16}	34.01 _{1.56}	62.05 _{0.94}	58.09 _{2.20}	54.68 _{0.05}	84.65 _{7.09}
FCBC	55.94 _{1.10}	25.69 _{0.73}	54.59 _{5.51}	63.86 _{4.72}	71.42 _{3.58}	51.28 _{2.86}	49.17 _{0.59}	25.84 _{0.56}	58.70 _{7.91}	54.55 _{3.52}	58.98 _{6.58}	74.33 _{13.48}
USPEC	57.09 _{1.73}	32.17 _{0.39}	64.68 _{3.04}	69.63 _{4.68}	83.08 _{2.35}	55.70 _{0.66}	54.50 _{0.26}	33.51 _{1.65}	63.30 _{0.42}	59.32 _{3.44}	52.52 _{0.50}	98.12 _{0.00}
EDCAG	56.03 _{1.96}	31.02 _{0.86}	55.10 _{2.17}	60.39 _{5.17}	71.70 _{1.63}	51.42 _{1.66}	46.12 _{1.76}	26.24 _{1.28}	53.41 _{4.97}	52.04 _{2.02}	61.75 _{11.94}	87.95 _{0.27}
LFCAG	61.25 _{6.79}	23.68 _{0.86}	71.84 _{1.98}	66.86 _{0.88}	67.83 _{0.67}	48.58 _{3.21}	41.86 _{1.82}	29.95 _{2.83}	61.57 _{4.39}	54.13 _{2.82}	59.76 _{4.77}	89.24 _{0.27}
FCAG	54.49 _{3.02}	31.29 _{0.97}	60.91 _{4.61}	73.74 _{2.07}	75.89 _{2.49}	50.87 _{1.55}	53.82 _{1.19}	30.87 _{1.44}	63.82 _{2.20}	56.56 _{6.62}	55.41 _{11.48}	87.17 _{6.03}
FAGC-R	59.36 _{1.13}	32.57 _{0.20}	67.93 _{2.05}	81.88 _{4.80}	85.45 _{0.98}	59.07 _{0.84}	53.97 _{0.84}	32.72 _{0.90}	63.24 _{1.36}	56.73 _{1.51}	55.26 _{0.02}	96.38 _{0.54}
CQP	52.90 _{4.04}	36.13 _{0.64}	51.65 _{7.44}	67.96 _{4.07}	89.35 _{1.67}	47.23 _{0.95}	54.53 _{0.42}	23.88 _{0.77}	45.23 _{3.31}	55.30 _{1.92}	34.81 _{0.22}	94.74 _{0.39}
FSSC	53.64 _{2.38}	19.10 _{1.44}	48.85 _{3.69}	54.49 _{7.81}	77.87 _{0.69}	50.72 _{0.96}	36.50 _{0.62}	20.54 _{1.96}	47.53 _{6.33}	47.80 _{6.75}	54.44 _{4.78}	73.24 _{3.76}
SC-FAGC(OURS)	63.09 _{0.00}	33.14 _{0.00}	78.33 _{0.00}	88.83 _{0.00}	86.55 _{0.00}	64.18 _{0.00}	55.93 _{0.00}	39.73 _{0.00}	64.81 _{0.00}	61.77 _{0.00}	73.27 _{0.00}	98.12 _{0.00}

in the number of neighbors. When the number of anchor points is large, the impact of changes in the k value on accuracy gradually diminishes. Meanwhile, when the number of neighbors remains constant, the clustering accuracy shows a trend of first increasing and then decreasing with an increase in the number of anchor points. This is because an increase in the number of anchor points makes the data information more comprehensive, but too many anchor points may introduce redundant or noisy information.

The fast convergence can be attributed to two main factors. First, the entropy regularization in our formulation provides a smooth optimization landscape, facilitating efficient gradient-based updates. Second, the Sinkhorn-Knopp algorithm employed in solving the DB-OT subproblem is known for its fast linear convergence under entropic regularization. These results validate that SC-FAGC achieves both computational efficiency and optimization stability, making it suitable for large-scale clustering tasks.

3.5. Cluster Size Constraint Verification

To evaluate cluster sizes at the data point level from the anchor-level assignments, we employ the Anchor-based Label Transmission (ALT) mechanism (Nie et al., 2024c). Specifically, let $Y \in \mathbb{R}^{n \times c}$ denote the label matrix of the original data points. Given the anchor assignment matrix $U \in \mathbb{R}^{m \times c}$ and the bipartite affinity matrix $B \in \mathbb{R}^{n \times m}$, the data-level labels can be obtained via $Y = BU$. The DB-OT solver enforces the bilateral constraints exactly on U , while data-level cluster sizes are induced through ALT. Their masses satisfy

$$(Y^\top \mathbf{1}_n)_j = \sum_{k=1}^m w_k u_{kj}, \quad w_k = \sum_{i=1}^n b_{ik}, \quad (11)$$

where w_k is the total data weight assigned to anchor k . When BKHK produces approximately balanced anchor regions, w_k is close to n/m , so the relative anchor-level bounds are approximately propagated to the data level. We

use this as an empirical approximation, not a hard assumption; the resulting data-level size behavior is checked directly in Fig. 4 and Fig. 7. This is an induced size-control mechanism rather than a hard data-level feasibility guarantee.

To validate the practical effect of bilateral cardinality control, we visualize the distribution of data-level cluster sizes obtained by our method on all twelve real-world datasets. At optimization time, the anchor-level lower bound α and upper bound β are computed as $\alpha = \lfloor 0.4 * m/c \rfloor$ and $\beta = \lceil 2.0 * m/c \rceil$, where m is the number of anchors and c is the number of clusters. The results are illustrated in Fig. 4, where the red dotted and dashed lines denote the corresponding propagated reference interval at the data level after ALT, and the blue solid line represents the ideal balanced cluster.

We further conduct a bound-sensitivity ablation for the bilateral constraints by testing five regimes relative to the average anchor-cluster size m/c : Very tight (0.8, 1.2), Tight (0.6, 1.5), Default (0.4, 2.0), Loose (0.2, 3.0), and Unconstrained. Overall, SC-FAGC is robust within a broad constrained range: performance changes are generally small among the first four settings, while removing constraints can degrade results on ambiguity-prone datasets. For example, on MSRA25 the constrained settings remain stable at 63.09 ACC, whereas the unconstrained variant drops to 56.09. On USPS20, constrained settings also outperform the unconstrained counterpart (e.g., 82.79 vs. 79.45). By contrast, on several balanced or well-separated datasets, the differences are marginal, indicating that the optimal solution already lies within the feasible interval once bilateral constraints are enabled. These observations support using (0.4, 2.0) as a practical default, with tighter bounds for near-balanced data and looser bounds for long-tail data. The complete ablation results are reported in Appendix C.

As shown in Fig. 4, the cluster sizes produced by SC-FAGC are generally concentrated within or near the propagated reference interval across datasets. For balanced datasets

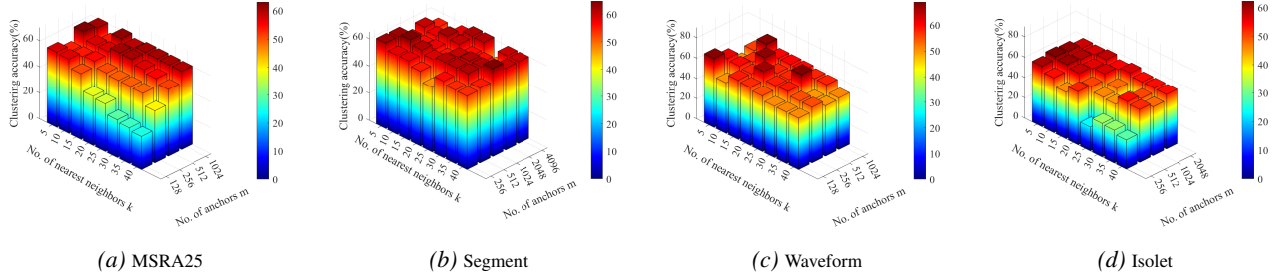


Figure 2. Sensitivity analysis of clustering accuracy w.r.t. number of anchors and neighbors on representative datasets. Additional results on other datasets are provided in Fig. 6 in the Appendix.

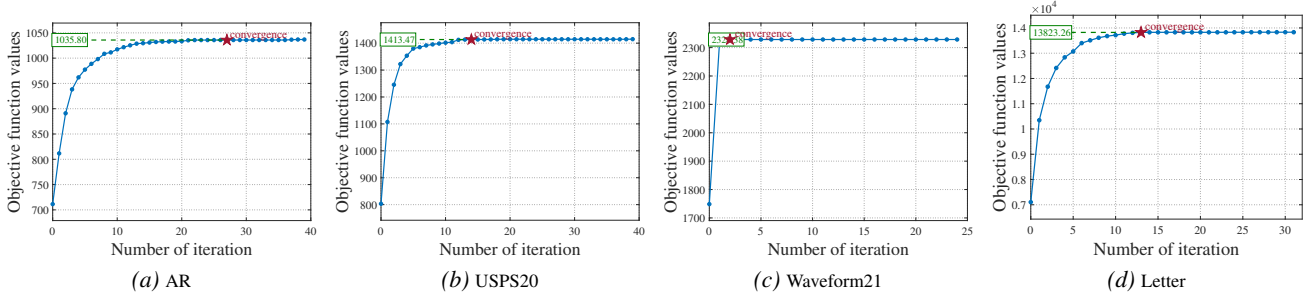


Figure 3. Convergence curves on representative datasets. Additional results on other datasets are provided in Fig. 8 in the Appendix.

such as Jaffe, AR, and Palm25, where each class contains approximately the same number of samples, the resulting cluster sizes are tightly distributed around the ideal balanced line. This demonstrates that SC-FAGC can recover balanced partitions when the underlying data distribution is close to uniform.

For imbalanced datasets such as MSRA25, Segment, and Waveform21, where class sizes vary significantly, SC-FAGC adaptively adjusts the cluster sizes and avoids severe collapse or dominance. In these cases, the flexibility provided by the bilateral constraints allows the algorithm to capture moderate natural imbalance while retaining practical size control after propagation.

These results support the core design principle of SC-FAGC: the Double-Bounded Optimal Transport formulation enforces size constraints exactly at the anchor level, and this control transfers effectively to data-level partitions through ALT in practice. This property is particularly valuable in practical applications where cluster balance is desired but strict equality is unrealistic.

3.6. Execution Time

To compare the time efficiency of the proposed method, we conduct runtime comparison experiments on BKNC, FRWL, FSC, FCBC, USPEC, EDCAG, LFCAG, FCAG, and FAGC-R. Since these methods are all based on anchors (except BKNC which operates on the original data), we exclude the time for anchor selection and bipartite graph

construction processes, focusing solely on the runtime of each algorithm, including initialization, optimization iterations, and discretization. Since the methods for anchor selection and bipartite graph construction are flexible and can involve multiple approaches, we do not consider their influence here. We set the convergence criteria for each algorithm according to the settings specified in their respective papers or source code. Each algorithm is run 10 times under the same number of anchor points and nearest neighbors, and the average running time along with the standard deviation is reported.

The runtime benchmark in Table 10 is therefore an anchor-based optimization comparison under a unified protocol; direct comparisons with FSSC are better viewed as complementary cross-paradigm evidence, since FSSC follows a different nearly-linear unconstrained spectral pipeline.

The comparison results of runtime are presented in Table 10 in Appendix J. From the table, several observations can be made. First, the proposed SC-FAGC method exhibits significant advantages in terms of runtime on most datasets. In particular, SC-FAGC achieves the fastest execution time on MSRA25, MNIST10, USPS20, Palm25, Mpeg7, Segment, Isolet, and Jaffe. This demonstrates the computational efficiency of our approach.

Second, compared to BKNC and LFCAG, SC-FAGC is substantially faster. BKNC operates on the original data without anchor acceleration, resulting in high computational costs especially on large datasets like MNIST10 (29.02s)

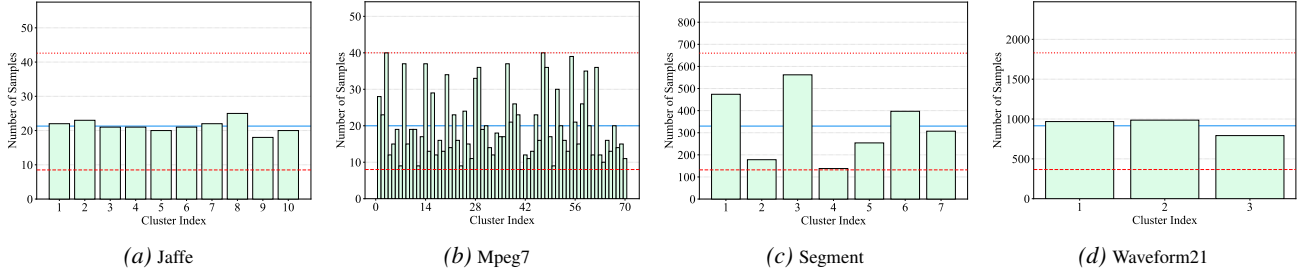


Figure 4. Distribution of data-level cluster sizes on representative real-world datasets. The red dotted and dashed lines indicate the propagated upper/lower reference interval after ALT, and the blue solid line represents the ideal balanced cluster. Additional results are provided in Fig. 7.

and COIL100 (65.89s). LFCAG, while anchor-based, involves iterative optimization with a regularization parameter search, leading to extremely long runtimes such as 709.79s on Letter and 472.08s on COIL100.

Third, SC-FAGC is competitive with efficient methods like USPEC and FSC while achieving significantly better clustering accuracy. For instance, on COIL100, SC-FAGC (6.78s) is comparable to USPEC (6.76s) but achieves 64.18% accuracy versus 55.70% for USPEC. On Isolet, SC-FAGC (0.95s) is much faster than FSC (5.42s) and USPEC (11.71s) while also delivering superior clustering performance.

The efficiency of SC-FAGC can be attributed to two key factors. First, by formulating the clustering problem as a Double-Bounded Optimal Transport problem, we leverage the Sinkhorn-Knopp algorithm which enjoys fast linear convergence under entropic regularization. Second, the anchor-based formulation reduces the problem size from n to m , where $m \ll n$, enabling efficient optimization on the anchor graph rather than the full data graph. As observed in the clustering performance results, SC-FAGC achieves state-of-the-art accuracy while maintaining competitive runtime, demonstrating a favorable balance between performance and computational complexity.

4. Conclusion

In this paper, we addressed the challenge of large-scale graph clustering with controllable cluster sizes, which is crucial for many real-world applications but often overlooked by existing methods. We proposed SC-FAGC (Size-Constrained Fast Anchor Graph Clustering), a unified framework that integrates anchor-based spectral clustering with bilateral cardinality constraints and entropy regularization.

The key contributions of our work are threefold. First, we formulated the size-constrained anchor graph clustering problem as a non-convex optimization problem with explicit lower and upper bounds on anchor-level cluster sizes, preventing degenerate solutions such as cluster collapse or excessive imbalance. Second, we developed an effi-

cient Iteratively Re-weighted (IRW) optimization algorithm that transforms the non-convex problem into a sequence of Double-Bounded Optimal Transport (DB-OT) subproblems, each solvable via the generalized Sinkhorn-Knopp algorithm with guaranteed convergence. Third, by operating on the anchor graph rather than the full data graph, SC-FAGC achieves $O(nm^2)$ complexity, making it scalable to large-scale datasets.

Extensive experiments on four synthetic datasets and twelve real-world benchmarks demonstrated the effectiveness of SC-FAGC. On synthetic datasets with complex geometric structures, SC-FAGC achieved near-perfect clustering accuracy, successfully capturing non-convex manifolds that challenge traditional methods. On real-world datasets, SC-FAGC consistently outperformed fourteen strong baseline methods across six evaluation metrics (ACC, NMI, Purity, Precision, Recall, and ARI), while strictly enforcing anchor-level size constraints and providing effective data-level size control after ALT. The runtime analysis further confirmed that SC-FAGC achieves a favorable trade-off between clustering performance and computational efficiency.

Future work will explore several promising directions. First, we plan to extend SC-FAGC to handle streaming data and online clustering scenarios. Second, incorporating deep learning feature extractors as a preprocessing step may further improve performance on high-dimensional and complex data. Third, investigating adaptive mechanisms for automatically determining the size constraint bounds based on data characteristics remains an interesting open problem. Finally, because the final clustering quality is still affected by anchor generation and bipartite graph construction, jointly optimizing BKHK anchor selection and the construction of B is an important limitation and future direction.

Impact Statement

This paper presents work whose goal is to advance the field of machine learning, specifically scalable graph clustering with controllable cluster sizes. The proposed method is a general-purpose unsupervised technique evaluated on stan-

dard public benchmarks, and we do not foresee direct negative societal consequences specific to it. As with any clustering method, downstream applications could in principle involve sensitive data (e.g., grouping individuals), so practitioners should ensure that the size constraints and resulting partitions are used in a manner consistent with fairness and privacy requirements in their specific deployment context. We do not anticipate other ethical concerns beyond those generally associated with unsupervised learning.

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Appendix

A. Discussion of Full-Graph Min-Cut Formulations

Recent full-graph min-cut formulations such as DNF (Xie et al., 2025c) study size-aware spectral clustering directly on the original graph. This line is closely related to SC-FAGC because both aim to avoid degenerate or severely imbalanced min-cut solutions through explicit size control. The main distinction is computational: DNF operates on the full graph, whereas SC-FAGC transfers the constrained optimization to an anchor-level graph and solves the resulting double-bounded optimal transport subproblems with Sinkhorn-type scaling. Thus, SC-FAGC targets the same practical need for controllable cluster sizes while preserving the scalability benefits of anchor-based spectral clustering.

B. Detailed Clustering Performance Metrics

This section reports the complete experimental results for NMI, Purity, Precision, Recall, and ARI on all twelve real-world datasets. These results serve as a supplement to the clustering Accuracy (ACC) typically reported in the main text (Table 3). The results consistently demonstrate that SC-FAGC achieves superior or competitive performance across all metrics against fourteen baseline methods while strictly adhering to the size constraints.

Table 4. Clustering NMI (Mean $\pm$ Std(%)) of Compared Methods on Real-World Datasets

Method	MSRA25	AR	Mnist10	USPS20	Palm25	COIL100	Mpeg7	Letter	Segment	Isolet	Waveform21	Jaffe
KM	55.66 _{4.14}	57.25 _{0.87}	51.85 _{1.39}	62.27 _{1.97}	88.84 _{1.12}	72.89 _{0.72}	68.52 _{1.13}	35.76 _{0.56}	56.19 _{2.98}	71.22 _{1.16}	35.75 _{0.00}	82.85 _{4.81}
NCut	67.77 _{2.63}	64.29 _{0.67}	71.65 _{1.79}	75.83 _{2.73}	90.77 _{0.75}	79.87 _{1.07}	70.98 _{0.53}	43.55 _{0.97}	53.00 _{3.89}	73.39 _{1.61}	36.48 _{0.00}	90.66 _{4.68}
BKNC	57.89 _{0.77}	62.98 _{0.40}	44.93 _{0.49}	60.48 _{2.01}	92.35 _{0.34}	69.06 _{0.46}	73.55 _{0.40}	30.39 _{0.87}	48.49 _{1.86}	61.64 _{1.03}	32.09 _{1.91}	92.48 _{1.14}
LSC	66.34 _{1.37}	63.61 _{0.63}	71.10 _{1.38}	76.10 _{2.31}	91.37 _{0.53}	73.98 _{0.40}	72.29 _{0.43}	43.57 _{0.99}	58.11 _{1.25}	73.19 _{0.80}	36.97 _{0.93}	92.81 _{2.84}
FRWL	68.38 _{1.62}	62.76 _{0.49}	64.73 _{1.71}	75.77 _{2.12}	92.92 _{0.26}	77.03 _{0.41}	71.74 _{0.81}	43.72 _{1.13}	59.32 _{0.54}	74.03 _{0.96}	37.74 _{0.00}	93.85 _{0.47}
FSC	70.29 _{1.49}	62.80 _{0.54}	65.90 _{2.16}	75.08 _{5.21}	91.67 _{0.50}	79.24 _{0.50}	72.19 _{1.41}	45.23 _{1.03}	57.17 _{1.22}	73.61 _{1.28}	36.90 _{0.01}	89.78 _{3.89}
FCBC	62.83 _{0.85}	53.93 _{0.64}	48.95 _{2.36}	59.32 _{3.76}	90.24 _{0.96}	76.29 _{0.79}	69.29 _{0.55}	34.79 _{0.35}	56.01 _{7.96}	70.91 _{0.67}	26.83 _{4.96}	80.29 _{8.41}
USPEC	64.80 _{1.95}	63.38 _{1.37}	63.30 _{0.52}	73.22 _{1.68}	93.75 _{0.45}	79.13 _{0.40}	71.09 _{0.78}	43.60 _{0.92}	59.28 _{0.51}	74.20 _{1.29}	36.65 _{0.04}	97.31 _{0.00}
EDCAG	66.00 _{1.18}	59.84 _{0.63}	52.98 _{0.61}	57.48 _{3.43}	90.62 _{0.65}	73.86 _{0.22}	63.76 _{0.79}	35.82 _{1.11}	48.34 _{5.30}	69.17 _{1.77}	31.93 _{18.79}	87.95 _{0.23}
LFCAG	54.35 _{4.41}	50.92 _{0.24}	68.12 _{0.07}	64.07 _{2.91}	87.80 _{1.28}	69.02 _{0.00}	60.68 _{0.85}	42.87 _{0.82}	55.87 _{1.06}	71.59 _{1.50}	37.38 _{2.33}	90.69 _{3.90}
FCAG	65.58 _{3.14}	62.98 _{0.40}	62.84 _{3.06}	75.33 _{1.45}	92.04 _{0.90}	77.35 _{0.62}	72.22 _{0.42}	42.73 _{1.05}	59.13 _{2.31}	74.46 _{1.36}	37.92 _{1.82}	90.94 _{2.98}
FAGC-R	70.94 _{1.91}	62.47 _{0.67}	64.70 _{1.67}	75.41 _{1.90}	95.13 _{0.54}	80.31 _{0.23}	71.82 _{0.46}	44.51 _{0.74}	60.59 _{0.68}	72.70 _{1.47}	36.91 _{0.00}	95.74 _{0.83}
CQP	62.55 _{1.72}	64.95 _{0.33}	47.63 _{6.13}	61.17 _{3.63}	95.65 _{0.58}	70.16 _{0.42}	71.79 _{0.21}	29.99 _{0.95}	33.61 _{2.38}	66.74 _{1.12}	0.07 _{0.02}	92.04 _{0.62}
FSSC	68.25 _{2.67}	51.02 _{1.45}	53.25 _{3.58}	66.43 _{3.77}	91.79 _{0.47}	75.04 _{0.56}	57.77 _{0.80}	26.00 _{0.55}	40.48 _{4.57}	68.26 _{2.92}	31.41 _{4.47}	80.95 _{1.92}
SC-FAGC(OURS)	69.63 _{0.00}	63.73 _{0.00}	73.91 _{0.00}	79.56 _{0.00}	95.77 _{0.00}	83.49 _{0.00}	73.33 _{0.00}	50.14 _{0.00}	61.17 _{0.00}	76.12 _{0.00}	45.16 _{0.00}	97.31 _{0.00}

Table 5. Clustering Purity (Mean $\pm$ Std(%)) of Compared Methods on Real-World Datasets

Method	MSRA25	AR	Mnist10	USPS20	Palm25	COIL100	Mpeg7	Letter	Segment	Isolet	Waveform21	Jaffe
KM	52.02 _{3.99}	28.65 _{1.03}	59.45 _{1.93}	71.17 _{3.21}	73.85 _{2.76}	49.39 _{1.59}	50.88 _{1.79}	28.11 _{0.64}	63.56 _{3.89}	58.67 _{2.29}	53.33 _{0.02}	80.66 _{6.81}
NCut	58.46 _{2.95}	35.70 _{1.49}	74.60 _{3.51}	79.09 _{1.18}	78.48 _{1.77}	59.44 _{1.93}	55.57 _{0.96}	35.73 _{0.89}	58.79 _{4.84}	58.53 _{2.66}	52.37 _{0.00}	87.65 _{7.23}
BKNC	57.12 _{1.41}	32.81 _{1.01}	55.40 _{1.22}	71.61 _{3.63}	87.17 _{0.71}	55.13 _{0.74}	59.79 _{0.40}	29.16 _{1.77}	58.10 _{1.25}	53.49 _{1.34}	58.52 _{5.11}	94.46 _{1.02}
LSC	60.99 _{1.25}	35.56 _{0.81}	74.21 _{2.70}	81.80 _{2.79}	79.57 _{1.26}	53.85 _{0.49}	57.89 _{0.65}	35.66 _{0.95}	67.73 _{2.55}	60.65 _{1.75}	53.92 _{2.72}	93.62 _{3.92}
FRWL	62.87 _{2.75}	34.12 _{0.58}	70.29 _{2.25}	81.73 _{2.16}	81.40 _{0.58}	57.66 _{0.82}	56.17 _{1.94}	34.17 _{1.13}	62.12 _{0.41}	61.42 _{1.21}	59.43 _{0.00}	95.93 _{0.27}
FSC	61.77 _{2.38}	33.56 _{0.81}	69.76 _{3.43}	79.60 _{7.24}	79.72 _{1.14}	58.98 _{0.35}	56.47 _{2.20}	36.35 _{1.17}	64.16 _{0.43}	61.01 _{1.62}	54.68 _{0.05}	87.15 _{5.62}
FCBC	58.50 _{2.26}	27.74 _{0.95}	57.62 _{3.59}	67.35 _{5.96}	76.93 _{2.11}	55.98 _{2.15}	53.14 _{0.58}	27.73 _{0.75}	62.41 _{4.46}	59.03 _{1.86}	60.92 _{4.56}	77.46 _{10.61}
USPEC	58.57 _{1.65}	31.26 _{0.51}	68.15 _{1.06}	78.44 _{3.03}	85.78 _{1.53}	61.31 _{1.34}	57.69 _{0.61}	36.15 _{1.46}	65.05 _{0.35}	61.90 _{2.71}	52.52 _{0.50}	98.12 _{0.00}
EDCAG	59.42 _{2.51}	32.31 _{0.97}	58.48 _{0.50}	63.56 _{3.85}	76.25 _{0.69}	55.18 _{1.24}	48.33 _{2.00}	27.98 _{1.65}	55.51 _{6.09}	55.43 _{2.19}	61.75 _{11.94}	87.95 _{0.27}
LFCAG	62.26 _{0.04}	25.32 _{0.96}	72.83 _{2.04}	70.31 _{2.71}	69.83 _{2.44}	51.48 _{3.09}	44.12 _{1.48}	31.85 _{2.24}	63.32 _{3.92}	57.36 _{1.13}	59.76 _{4.77}	91.71 _{5.44}
FCAG	57.33 _{2.80}	32.81 _{1.01}	66.68 _{3.57}	81.04 _{2.26}	80.28 _{2.11}	55.20 _{1.37}	56.46 _{1.18}	33.17 _{1.25}	65.84 _{2.53}	60.84 _{2.71}	56.68 _{6.13}	88.90 _{4.83}
FAGC-R	62.58 _{1.83}	33.63 _{0.07}	70.17 _{2.39}	83.09 _{3.34}	87.05 _{1.38}	63.04 _{0.60}	57.42 _{0.77}	34.60 _{0.59}	64.32 _{1.18}	60.39 _{0.96}	55.26 _{0.02}	96.38 _{0.54}
CQP	56.14 _{3.96}	37.34 _{0.71}	52.64 _{7.09}	69.40 _{4.16}	89.75 _{1.44}	50.38 _{0.94}	58.24 _{0.27}	25.53 _{0.46}	46.71 _{3.04}	57.77 _{1.55}	35.27 _{0.27}	94.74 _{0.39}
FSSC	58.20 _{3.09}	20.76 _{1.54}	55.91 _{5.59}	68.34 _{4.84}	80.78 _{0.37}	58.88 _{1.50}	42.02 _{0.77}	22.12 _{1.90}	49.77 _{7.07}	53.80 _{5.42}	56.31 _{3.99}	77.31 _{2.67}
SC-FAGC(OURS)	63.09 _{0.00}	34.79 _{0.00}	80.70 _{0.00}	88.83 _{0.00}	88.80 _{0.00}	67.21 _{0.00}	59.00 _{0.00}	41.42 _{0.00}	66.02 _{0.00}	63.23 _{0.00}	73.27 _{0.00}	98.12 _{0.00}

C. Bound-Sensitivity Ablation for Bilateral Constraints

This appendix reports the complete clustering ACC (%) under different bilateral cardinality constraint settings. The five settings are defined relative to the average anchor-cluster size m/c . Overall, SC-FAGC remains stable across constrained regimes, while the unconstrained variant can degrade on ambiguity-prone datasets.

SC-FAGC: Size Constrained Fast Anchor Graph Clustering

Table 6. Clustering Precision (Mean $\pm$ Std(%)) of Compared Methods on Real-World Datasets

Method	MSRA25	AR	Mnist10	USPS20	Palm25	COIL100	Mpeg7	Letter	Segment	Isolet	Waveform21	Jaffe
KM	34.80 _{5.35}	10.60 _{0.78}	43.62 _{1.46}	57.29 _{2.71}	57.01 _{3.86}	35.50 _{1.67}	27.63 _{2.45}	15.72 _{0.60}	51.86 _{3.64}	45.51 _{3.17}	50.01 _{0.00}	68.34 _{9.29}
NCut	41.25 _{3.92}	18.32 _{1.01}	62.00 _{4.52}	67.00 _{6.22}	63.99 _{3.02}	36.61 _{3.99}	36.07 _{1.31}	16.81 _{1.78}	43.51 _{3.87}	46.45 _{4.04}	50.07 _{0.00}	78.41 _{11.29}
BKNC	43.59 _{0.79}	10.11 _{0.57}	39.34 _{0.43}	57.70 _{1.75}	78.68 _{1.20}	43.38 _{0.91}	41.99 _{0.69}	17.17 _{0.97}	46.52 _{1.79}	42.84 _{1.82}	50.53 _{1.90}	89.39 _{1.66}
LSC	44.98 _{2.89}	17.26 _{0.95}	63.32 _{3.64}	70.35 _{4.09}	65.12 _{2.26}	39.98 _{1.65}	36.21 _{1.51}	20.02 _{1.45}	55.08 _{1.56}	49.72 _{2.25}	50.44 _{0.89}	88.03 _{6.81}
FRWL	50.14 _{2.94}	15.99 _{0.83}	57.21 _{2.59}	71.59 _{3.55}	67.59 _{1.11}	45.36 _{1.08}	38.58 _{2.73}	22.02 _{1.84}	54.18 _{2.33}	52.26 _{3.58}	51.71 _{0.00}	92.18 _{0.54}
FSC	46.35 _{2.22}	15.82 _{0.91}	55.18 _{5.77}	70.13 _{10.92}	64.87 _{2.66}	44.08 _{1.28}	33.42 _{3.30}	20.52 _{1.65}	52.52 _{2.03}	49.40 _{2.92}	50.38 _{0.01}	77.43 _{9.57}
FCBC	44.85 _{1.15}	8.41 _{0.38}	42.24 _{2.88}	54.34 _{6.50}	61.39 _{4.15}	42.42 _{2.64}	30.04 _{0.88}	16.41 _{0.27}	46.39 _{10.29}	46.53 _{3.68}	48.97 _{4.09}	64.07 _{14.41}
USPEC	40.49 _{3.38}	16.21 _{2.04}	54.45 _{2.05}	67.06 _{3.93}	76.05 _{2.60}	45.21 _{0.40}	35.68 _{4.11}	17.98 _{1.35}	55.77 _{0.69}	52.26 _{2.82}	50.10 _{0.04}	96.05 _{0.00}
EDCAG	44.82 _{2.35}	11.39 _{0.87}	41.98 _{1.73}	39.55 _{5.15}	63.20 _{2.19}	29.70 _{2.38}	17.11 _{1.34}	12.19 _{0.83}	43.56 _{3.92}	43.42 _{2.92}	50.48 _{11.36}	78.64 _{1.11}
LFCAG	53.13 _{3.77}	5.91 _{0.02}	63.78 _{0.40}	58.86 _{0.74}	53.47 _{1.24}	37.39 _{4.13}	27.03 _{1.26}	20.71 _{1.86}	53.66 _{2.32}	46.23 _{3.42}	51.96 _{1.85}	86.12 _{6.81}
FCAG	41.90 _{5.62}	16.18 _{0.67}	51.47 _{4.29}	70.80 _{2.31}	66.58 _{2.86}	43.91 _{1.55}	41.39 _{1.34}	20.89 _{1.28}	55.87 _{2.69}	49.78 _{2.93}	51.51 _{3.46}	80.27 _{7.74}
FAGC-R	49.13 _{3.10}	17.07 _{0.93}	58.49 _{1.58}	73.95 _{4.78}	78.17 _{3.42}	47.38 _{0.59}	39.64 _{1.58}	22.44 _{1.18}	55.36 _{0.85}	49.69 _{1.74}	50.56 _{0.00}	92.58 _{1.08}
CQP	47.16 _{3.07}	21.37 _{0.38}	40.12 _{6.84}	56.71 _{4.00}	87.16 _{1.66}	40.21 _{0.60}	41.70 _{0.39}	14.64 _{0.49}	33.54 _{1.53}	47.85 _{2.04}	33.35 _{0.01}	89.98 _{0.75}
FSSC	44.03 _{4.73}	6.88 _{0.87}	36.20 _{3.56}	51.04 _{6.61}	64.20 _{2.03}	27.52 _{2.27}	10.77 _{1.76}	6.87 _{0.19}	33.99 _{2.02}	33.69 _{2.27}	49.68 _{1.42}	59.32 _{3.59}
SC-FAGC(OURS)	43.98 _{0.00}	16.74 _{0.00}	68.29 _{0.00}	81.10 _{0.00}	79.85 _{0.00}	54.86 _{0.00}	34.22 _{0.00}	22.60 _{0.00}	51.07 _{0.00}	53.90 _{0.00}	60.80 _{0.00}	96.05 _{0.00}

Table 7. Clustering Recall (Mean $\pm$ Std(%)) of Compared Methods on Real-World Datasets

Method	MSRA25	AR	Mnist10	USPS20	Palm25	COIL100	Mpeg7	Letter	Segment	Isolet	Waveform21	Jaffe
KM	44.40 _{3.10}	17.04 _{0.80}	46.86 _{1.59}	58.13 _{2.62}	73.89 _{2.32}	49.71 _{1.42}	45.18 _{1.51}	18.36 _{0.52}	59.48 _{2.94}	54.78 _{2.34}	50.38 _{0.00}	80.91 _{5.76}
NCut	56.70 _{1.11}	23.55 _{0.50}	71.65 _{3.07}	71.55 _{4.86}	79.10 _{1.58}	62.29 _{1.60}	42.52 _{0.75}	30.05 _{1.12}	62.44 _{2.83}	59.47 _{2.48}	50.21 _{0.00}	93.25 _{2.87}
BKNC	45.16 _{1.48}	18.58 _{0.21}	40.41 _{0.67}	58.80 _{2.18}	88.86 _{1.31}	55.52 _{0.67}	46.20 _{0.20}	18.79 _{1.25}	53.21 _{2.41}	49.73 _{0.62}	50.85 _{2.20}	89.87 _{1.34}
LSC	52.61 _{1.94}	22.54 _{0.33}	68.32 _{2.52}	69.74 _{7.76}	82.20 _{1.00}	47.60 _{8.83}	45.80 _{7.70}	26.45 _{1.00}	60.12 _{1.71}	56.03 _{1.65}	50.95 _{0.95}	91.61 _{2.70}
FRWL	54.39 _{1.89}	20.99 _{0.26}	62.11 _{3.29}	71.00 _{5.62}	86.54 _{1.04}	52.28 _{1.21}	45.98 _{0.36}	26.18 _{0.94}	60.33 _{0.37}	58.50 _{1.57}	52.41 _{0.00}	92.20 _{0.56}
FSC	57.19 _{0.92}	22.15 _{0.48}	65.25 _{3.77}	73.40 _{8.56}	81.94 _{0.93}	56.87 _{1.11}	48.83 _{0.67}	29.02 _{1.44}	61.47 _{1.13}	58.15 _{0.79}	50.53 _{0.01}	91.81 _{2.29}
FCBC	48.40 _{2.03}	15.09 _{1.30}	44.88 _{3.67}	55.87 _{4.83}	76.40 _{1.93}	54.17 _{1.08}	45.21 _{0.15}	17.75 _{0.35}	63.22 _{6.81}	54.56 _{5.84}	58.31 _{5.50}	77.78 _{8.64}
USPEC	52.94 _{1.29}	20.77 _{0.92}	59.18 _{1.25}	68.96 _{3.77}	84.30 _{1.05}	54.64 _{1.33}	47.59 _{0.80}	27.98 _{2.06}	63.32 _{0.92}	57.40 _{1.54}	50.22 _{0.04}	96.58 _{0.00}
EDCAG	51.39 _{2.34}	18.42 _{0.76}	47.28 _{1.29}	49.25 _{3.03}	75.14 _{1.92}	50.14 _{1.39}	41.40 _{1.20}	20.57 _{1.59}	47.80 _{3.21}	48.12 _{3.11}	51.86 _{0.62}	82.67 _{0.27}
LFCAG	54.85 _{4.49}	16.04 _{0.32}	64.73 _{0.76}	58.85 _{1.83}	77.06 _{2.35}	53.75 _{0.88}	37.62 _{0.95}	22.44 _{2.03}	55.11 _{1.30}	56.60 _{0.67}	51.97 _{1.89}	87.12 _{5.61}
FCAG	52.32 _{2.38}	20.38 _{0.66}	59.13 _{3.03}	70.16 _{1.34}	81.28 _{2.12}	52.44 _{1.38}	47.22 _{0.77}	23.51 _{1.15}	59.69 _{2.41}	60.78 _{2.32}	53.15 _{4.67}	91.59 _{3.08}
FAGC-R	57.40 _{0.76}	20.90 _{0.65}	62.23 _{2.32}	79.74 _{1.66}	88.70 _{0.88}	57.98 _{0.48}	45.19 _{0.52}	26.16 _{0.77}	61.61 _{2.79}	58.56 _{1.86}	50.88 _{0.00}	94.10 _{0.98}
CQP	45.99 _{3.00}	21.37 _{0.38}	40.01 _{6.82}	52.68 _{3.72}	87.16 _{1.66}	40.21 _{0.60}	41.70 _{0.39}	14.63 _{0.48}	33.54 _{1.53}	47.85 _{2.04}	33.31 _{0.01}	89.85 _{0.75}
FSSC	53.75 _{3.25}	12.22 _{0.87}	51.67 _{4.60}	60.49 _{2.85}	84.00 _{1.53}	55.26 _{2.08}	35.94 _{1.39}	32.04 _{1.56}	53.08 _{3.94}	61.37 _{3.58}	52.30 _{5.55}	79.84 _{0.63}
SC-FAGC(OURS)	55.74 _{0.00}	22.16 _{0.00}	78.33 _{0.00}	83.51 _{0.00}	89.75 _{0.00}	63.83 _{0.00}	46.74 _{0.00}	34.33 _{0.00}	64.76 _{0.00}	67.56 _{0.00}	61.29 _{0.00}	96.58 _{0.00}

Table 8. Clustering ARI (Mean $\pm$ Std(%)) of Compared Methods on Real-World Datasets

Method	MSRA25	AR	Mnist10	USPS20	Palm25	COIL100	Mpeg7	Letter	Segment	Isolet	Waveform21	Jaffe
KM	32.35 _{5.21}	12.05 _{0.76}	38.83 _{1.61}	52.58 _{2.88}	63.94 _{3.27}	40.73 _{1.46}	33.09 _{2.07}	13.34 _{0.49}	47.33 _{3.56}	47.45 _{2.49}	25.14 _{0.00}	70.88 _{8.57}
NCut	41.94 _{3.39}	19.76 _{0.79}	62.35 _{4.00}	65.26 _{5.86}	70.41 _{2.33}	45.32 _{3.35}	38.11 _{1.00}	17.41 _{1.72}	41.30 _{4.23}	49.98 _{2.83}	25.15 _{0.00}	83.09 _{8.79}
BKNC	39.09 _{1.08}	12.03 _{0.55}	33.07 _{0.59}	53.18 _{2.12}	82.45 _{1.01}	47.80 _{0.76}	43.18 _{0.47}	14.51 _{1.12}	40.62 _{2.40}	43.70 _{1.41}	25.91 _{2.98}	88.53 _{1.65}
LSC	43.26 _{2.53}	18.68 _{0.68}	61.73 _{3.51}	66.46 _{4.84}	72.36 _{1.61}	42.86 _{1.06}	39.50 _{0.95}	19.22 _{1.24}	50.06 _{1.78}	50.67 _{1.91}	25.85 _{1.36}	88.55 _{5.22}
FRWL	47.97 _{2.42}	17.28 _{0.64}	54.85 _{3.23}	67.86 _{5.03}	75.63 _{0.52}	48.02 _{1.14}	42.24 _{1.66}	20.59 _{1.62}	49.48 _{1.57}	53.29 _{2.87}	27.83 _{0.00}	91.36 _{0.61}
FSC	46.12 _{1.79}	17.56 _{0.77}	54.56 _{3.62}	68.13 _{10.94}	72.09 _{1.70}	49.10 _{1.23}	38.61 _{2.45}	20.40 _{1.46}	48.75 _{1.77}	51.35 _{2.83}	25.61 _{0.01}	81.85 _{7.45}
FCBC	41.38 _{1.60}	9.72 _{0.64}	37.02 _{3.57}	49.52 _{5.80}	67.70 _{3.07}	46.98 _{2.09}	35.04 _{0.68}	13.60 _{0.22}	43.81 _{10.02}	48.02 _{4.34}	26.35 _{5.42}	66.53 _{13.78}
USPEC	40.02 _{2.44}	18.01 _{1.68}	51.67 _{1.90}	64.07 _{3.93}	79.75 _{1.75}	48.92 _{0.34}	39.73 _{2.61}	18.02 _{1.31}	50.04 _{0.57}	52.80 _{2.27}	25.18 _{0.06}	95.92 _{0.00}
EDCAG	42.60 _{1.03}	13.07 _{0.87}	37.85 _{1.21}	36.23 _{5.21}	68.33 _{2.11}	36.45 _{1.60}	22.69 _{1.27}	10.98 _{0.73}	35.96 _{3.88}	43.37 _{3.13}	25.96 _{16.91}	78.49 _{0.66}
LFCAG	46.18 _{4.73}	7.38 _{0.06}	60.23 _{0.63}	53.92 _{1.39}	62.71 _{1.67}	35.37 _{3.92}	30.35 _{1.16}	18.27 _{2.02}	46.66 _{2.22}	43.37 _{3.13}	27.94 _{2.79}	85.18 _{6.90}
FCAG	40.72 _{4.82}	17.18 _{0.65}	49.59 _{4.08}	66.95 _{2.03}	72.91 _{2.60}	47.22 _{1.37}	43.29 _{0.99}	18.81 _{1.25}	50.40 _{2.88}	52.68 _{2.01}	27.94 _{5.87}	83.75 _{5.96}
FAGC-R	48.13 _{2.30}	17.94 _{0.77}	55.72 _{1.96}	73.75 _{3.51}	82.90 _{2.26}	51.62 _{0.49}	42.22 _{1.13}	20.87 _{0.95}	50.93 _{1.25}	51.68 _{1.15}	25.95 _{0.00}	92.62 _{1.14}
CQP	41.68 _{3.31}	20.64 _{0.38}	33.40 _{7.59}	49.40 _{4.30}	87.04 _{1.68}	39.61 _{0.61}	40.90 _{0.40}	11.22 _{0.50}	22.50 _{1.79}	45.77 _{2.12}	0.01 _{0.02}	88.85 _{0.83}
FSSC	42.99 _{4.24}	7.69 _{0.94}	34.86 _{4.37}	49.31 _{5.45}	72.47 _{1.54}	35.88 _{1.66}	14.71 _{2.01}	5.31 _{0.32}	29.09 _{2.73}	40.36 _{3.74}	25.49 _{3.52}	64.07 _{3.02}
SC-FAGC(OURS)	43.84 _{0.00}	18.21 _{0.00}	69.73 _{0.00}	80.13 _{0.00}	84.35 _{0.00}	58.57 _{0.00}	38.55 _{0.00}	23.72 _{0.00}	50.98 _{0.00}	58.18 _{0.00}	31.43 _{0.00}	95.92 _{0.00}

Table 9. Clustering ACC (%) under different bilateral cardinality constraint settings. The five settings correspond to Very tight (0.8, 1.2), Tight (0.6, 1.5), Default (0.4, 2.0), Loose (0.2, 3.0), and Unconstrained.

Setting	MSRA25	AR	Mnist10	USPS20	Palm25	COIL100	Mpeg7	Letter	Segment	Isolet	Waveform21	Jaffe
Very tight (0.8, 1.2)	63.09	33.14	74.90	82.79	83.35	59.85	55.21	38.65	64.63	61.68	74.65	98.12
Tight (0.6, 1.5)	63.09	33.14	75.44	82.74	83.35	59.85	55.21	38.67	64.63	61.68	74.65	98.12
Default (0.4, 2.0)	63.09	33.14	78.06	80.80	83.35	59.85	55.21	38.67	64.81	61.68	74.65	98.12
Loose (0.2, 3.0)	63.09	33.14	76.99	79.61	83.35	59.85	55.21	38.45	64.81	61.68	74.65	98.12
Unconstrained	56.09	33.01	77.00	79.45	83.35	59.36	55.50	38.38	64.81	61.68	74.65	98.12

D. Additional Synthetic Dataset Visualizations

This appendix provides additional visualization results on synthetic datasets, complementing Fig. 1 in the main text.

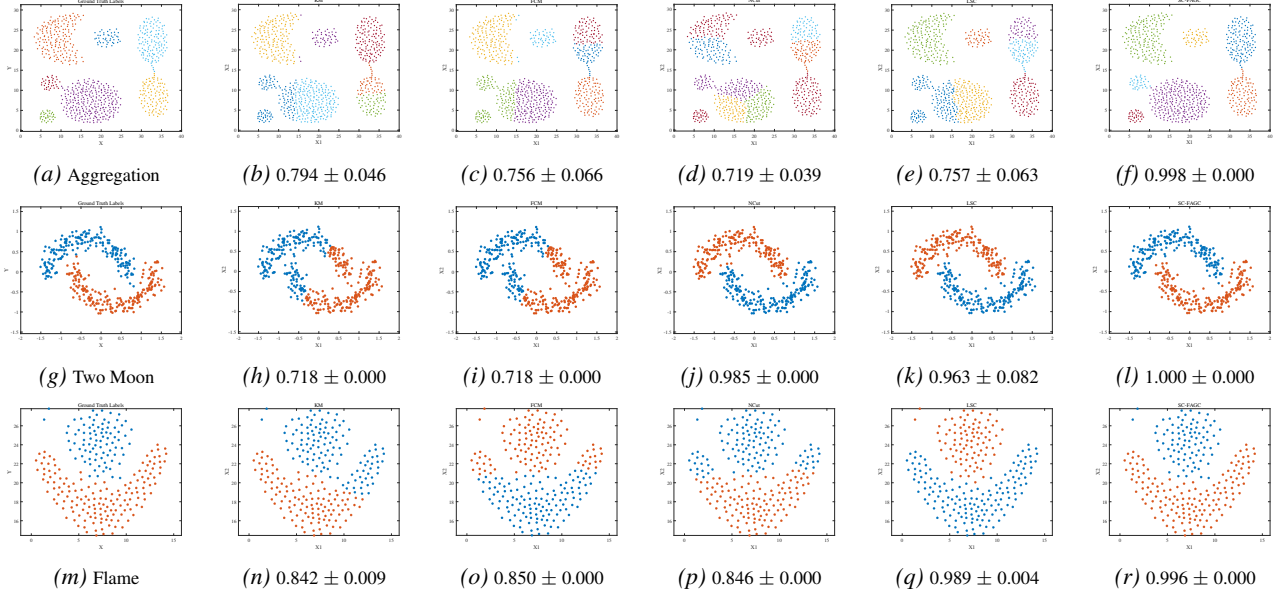


Figure 5. Visualization of clustering results on additional synthetic datasets (Aggregation, Two Moon, Flame). The first column shows Ground Truth, followed by KM, FCM, NCut, LSC, and SC-FAGC. Values denote mean ACC $\pm$ std.

E. Additional Sensitivity Analysis Results

This appendix provides additional sensitivity analysis results on the remaining eight datasets. These results complement Fig. 2 in the main text.

F. Additional Cluster Size Distribution Results

This appendix provides additional cluster size distribution results on the remaining eight datasets, complementing Fig. 4 in the main text.

G. Additional Convergence Curve Results

This appendix provides additional convergence curve results on the remaining eight datasets, complementing Fig. 3 in the main text.

H. Complexity Analysis

In this section, we provide a detailed analysis of the computational complexity of the proposed SC-FAGC framework. The overall process consists of three main stages: anchor generation, bipartite graph construction, and optimization.

1. Anchor Generation and Graph Construction. Let n be the number of data points, m the number of anchors, and d the dimensionality of the data.

- **Anchor Selection:** We use Balanced K-means based Hierarchical K-means (BKHK), which recursively bisects the data with a balanced 2-means procedure until m anchors are obtained. With recursion depth $\log_2 m$ and a fixed number of 2-means iterations per level, all n points are processed once per level, so anchor generation costs $O(nd \log m)$. Since $m \ll n$, this stage is linear in n up to the logarithmic anchor factor.
- **Bipartite Graph (B) Construction:** Calculating the distance between n samples and m anchors takes $O(nmd)$. For

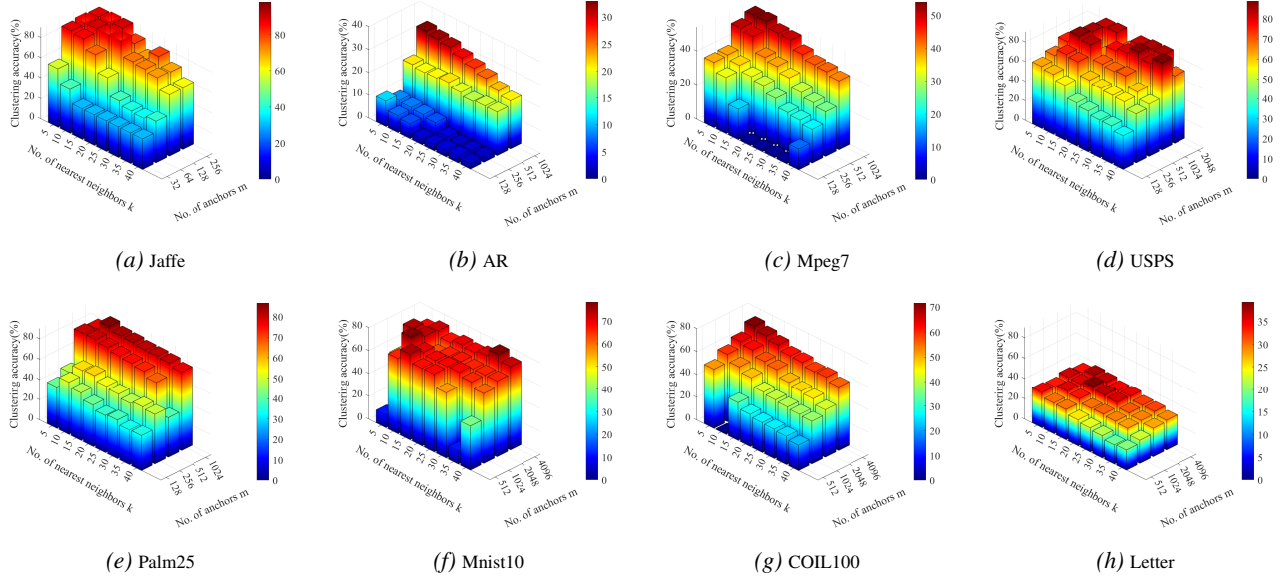


Figure 6. Sensitivity analysis of clustering accuracy w.r.t. number of anchors and neighbors on additional datasets (continued from Fig. 2).

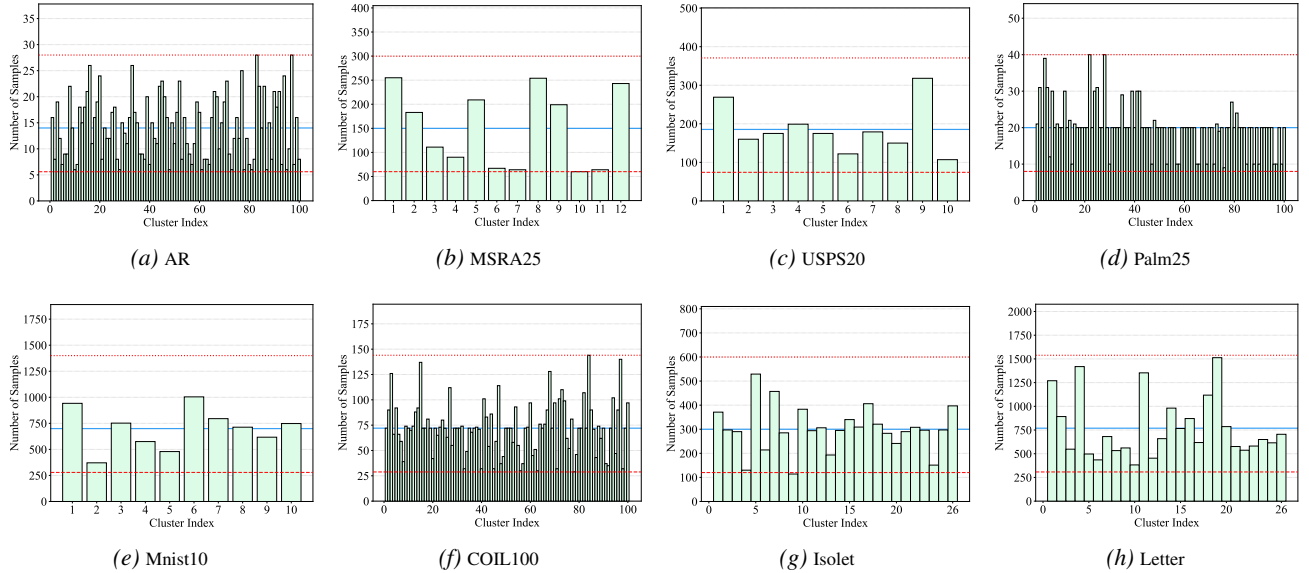


Figure 7. Distribution of data-level cluster sizes on additional real-world datasets (continued from Fig. 4). The red dotted and dashed lines indicate the propagated upper/lower reference interval after ALT, and the blue solid line represents the ideal balanced cluster.

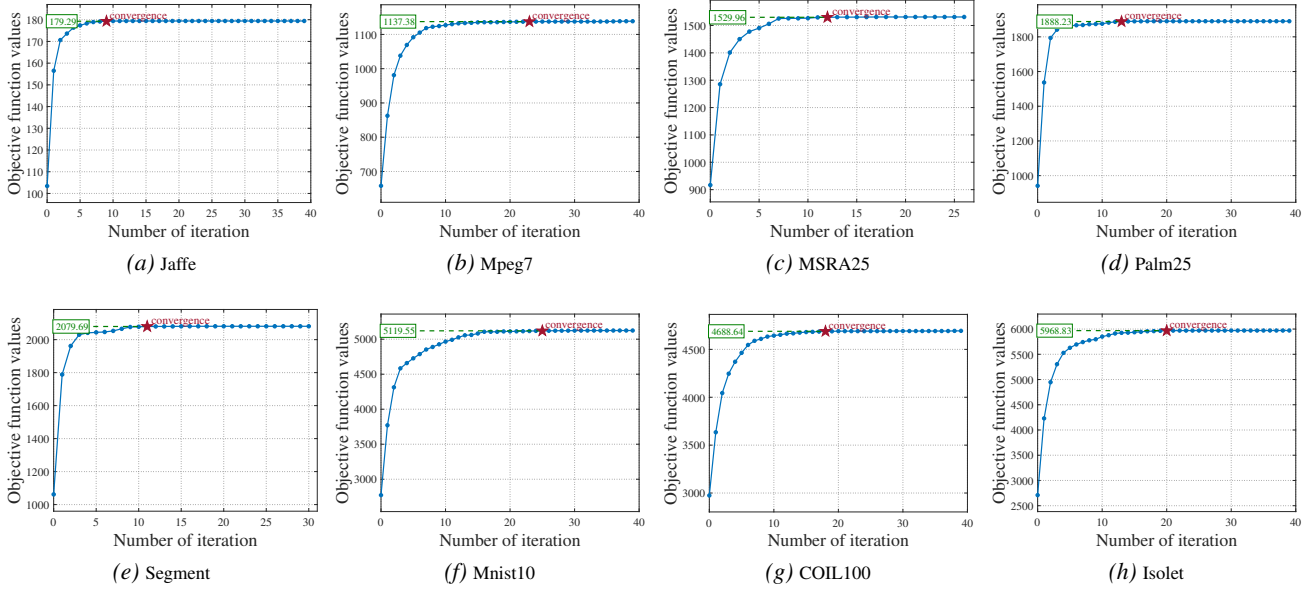


Figure 8. Convergence curves on additional datasets (continued from Fig. 3). The red star marks the convergence point.

each sample, sorting all anchor distances to find the s -nearest anchors takes $O(m \log m)$, yielding $O(nm \log m)$ in total. Since the distance computation dominates in the typical high-dimensional setting, constructing $B \in \mathbb{R}^{n \times m}$ has complexity $O(nmd)$.

- **Anchor Graph (A) Construction:** The anchor-level similarity matrix is defined as $A = B^\top B$. This involves multiplying an $m \times n$ matrix by an $n \times m$ matrix.
 - If B is treated as a dense matrix, the complexity is $O(nm^2)$.
 - If B is sparse with only s non-zero entries per row (where $s \ll m$), the multiplication can be implemented more efficiently in $O(ns^2)$.

In the worst-case (dense) scenario, this step is $O(nm^2)$.

2. Optimization via IRW and Sinkhorn-Knopp. The core optimization involves iteratively solving the Double-Bounded Optimal Transport (DB-OT) problem.

- **Gradient Computation:** In each iteration of the Iteratively Re-weighted (IRW) algorithm, we compute the gradient $D = 2AU$. Since $A \in \mathbb{R}^{m \times m}$ and $U \in \mathbb{R}^{m \times c}$, this matrix multiplication takes $O(m^2c)$.
- **Sinkhorn-Knopp Algorithm:** Solving the DB-OT subproblem requires iterative matrix-vector multiplications. Let T_{sink} be the number of Sinkhorn iterations. Each iteration involves operations on $m \times c$ matrices, with complexity $O(mc)$. Thus, one DB-OT solution takes $O(T_{\text{sink}}mc)$.
- **Total Optimization Cost:** Assuming the outer IRW loop runs for T iterations, the total optimization complexity is $O(T(m^2c + T_{\text{sink}}mc))$.

3. Final Assignment. Recovering the cluster labels for all n data points via $Y = BU$ takes $O(nmc)$ (or $O(nsc)$ for sparse B).

Summary. The total time complexity of SC-FAGC is:

$$O(nd \log m + nmd + nm^2 + T(m^2c + T_{\text{sink}}mc)). \quad (12)$$

Here, $nd \log m$ accounts for BKHK anchor generation, nmd for distance computation and bipartite graph construction, nm^2 for computing $A = B^\top B$, and $T(m^2c + T_{\text{sink}}mc)$ for the optimization loop. Given that $m \ll n$ and d, T, T_{sink} , and c are

treated as constants, the dominant terms are $O(nmd)$ and $O(nm^2)$, making the complexity linear with respect to the number of samples n , i.e., $O(n)$. This is a significant improvement over standard spectral clustering, which has a complexity of $O(n^3)$ due to the eigendecomposition of the full Laplacian matrix. Additionally, compared to some anchor-based methods that perform spectral analysis on A (requiring $O(m^3)$), SC-FAGC achieves competitive efficiency by avoiding explicit eigendecomposition for the final clustering result.

I. Convergence Analysis

To solve the non-convex optimization problem in Eq. (5), we employ the Iteratively Re-weighted (IRW) optimization framework (Nie et al., 2024c). In this section, we provide a theoretical proof of the convergence of our proposed algorithm.

Problem Reformulation. Our objective function is:

$$\max_{U \in \mathcal{C}} \mathcal{J}(U) = \text{Tr}(U^\top AU) + \epsilon H(U), \quad (13)$$

where $A = B^\top B$ is a positive semi-definite matrix (making $f(U) = \text{Tr}(U^\top AU)$ a convex function with respect to U) and $H(U)$ is the concave entropy function. Thus, the problem can be viewed as maximizing the sum of a convex function and a concave function.

General IRW Framework. According to the IRW framework (Nie et al., 2024c), for a general maximization problem of the form:

$$\max_{x \in \mathcal{C}} F(x) = f(x) + g(x), \quad (14)$$

where $f(x)$ is convex and $g(x)$ is concave, we can construct a surrogate function by linearizing the convex term $f(x)$ at the current point x_t . By the definition of convexity, we have the following lower bound for $f(x)$:

$$f(x) \geq f(x_t) + \langle \nabla f(x_t), x - x_t \rangle. \quad (15)$$

Substituting this into the objective, we define the surrogate function $Q(x, x_t)$ as:

$$Q(x, x_t) = f(x_t) + \langle \nabla f(x_t), x - x_t \rangle + g(x). \quad (16)$$

From the inequality above, it holds that $F(x) \geq Q(x, x_t)$ for all x , and $F(x_t) = Q(x_t, x_t)$.

Convergence of SC-FAGC. In our proposed algorithm, at iteration t , we update U by solving the subproblem:

$$U_{t+1} = \arg \max_{U \in \mathcal{C}} Q(U, U_t) = \arg \max_{U \in \mathcal{C}} \langle \nabla f(U_t), U \rangle + \epsilon H(U). \quad (17)$$

Here, the gradient is $\nabla f(U_t) = 2AU_t$, which corresponds to the matrix D in our algorithm (Step 1). The subproblem is equivalent to solving the Double-Bounded Optimal Transport problem (Step 2).

Since U_{t+1} maximizes the surrogate function $Q(U, U_t)$, we have:

$$Q(U_{t+1}, U_t) \geq Q(U_t, U_t) = F(U_t). \quad (18)$$

Combining this with the lower bound property $F(U_{t+1}) \geq Q(U_{t+1}, U_t)$, we obtain:

$$F(U_{t+1}) \geq F(U_t). \quad (19)$$

This inequality ensures that the objective function value matches or increases monotonically in each iteration. Since the objective function is bounded (due to the constraints on U), the algorithm is guaranteed to converge.

J. Execution Time Comparison

K. Derivation of the Closed-Form Anchor Assignment

We derive Eq. (3) for a fixed data point x_i . Let $d_{ij} = \|x_i - z_j\|_2^2$ and write b_j for b_{ij} . The assignment problem is

$$\min_{b^\top \mathbf{1}_m = 1, b \geq 0} \sum_{j=1}^m d_{ij} b_j + \gamma \sum_{j=1}^m b_j^2. \quad (20)$$

Table 10. Comparison of Runtime (Mean $\pm$ Std(%)) Across Different Methods on Various Datasets

Method	MSRA25	AR	MNIST10	USPS20	Palm25	COIL100	Mpeg7	Letter	Segment	Isotet	Waveform21	Jaffe
BKNC	4.3416 _{0.7001}	3.3844 _{0.0427}	29.0192 _{0.1397}	1.9010 _{0.0088}	10.8365 _{0.4196}	65.8859 _{2.5280}	9.4217 _{0.0576}	10.0655 _{0.0263}	0.3177 _{0.0071}	31.6990 _{0.2407}	0.2105 _{0.0140}	0.5800 _{0.0537}
FRWL	0.6351 _{0.0187}	4.2442 _{0.6621}	6.0947 _{1.0176}	1.5510 _{0.0631}	1.6787 _{0.0775}	11.4793 _{0.1636}	9.6634 _{0.3527}	8.8021 _{0.6616}	0.5851 _{0.0349}	6.9819 _{0.3376}	0.5445 _{0.0450}	0.2716 _{0.0091}
FSC	0.5580 _{0.0163}	2.9118 _{0.0297}	4.3869 _{0.3603}	0.9545 _{0.0135}	1.0219 _{0.0314}	8.2622 _{0.0559}	6.4166 _{0.0805}	7.1559 _{0.7485}	0.3489 _{0.0073}	5.4207 _{0.0568}	0.3935 _{0.0177}	0.2649 _{0.0152}
FCBC	0.9372 _{0.0488}	75.0684 _{4.9871}	11.0599 _{0.6695}	0.8150 _{0.0377}	9.1912 _{0.7461}	152.5069 _{9.9963}	166.7566 _{98.0794}	1.3503 _{0.1847}	0.0360 _{0.0060}	32.5070 _{0.9506}	0.0176 _{0.0015}	0.3239 _{0.0100}
USPEC	0.3976 _{0.0112}	2.3360 _{0.0534}	5.9546 _{0.2495}	0.8053 _{0.0704}	0.6774 _{0.0687}	6.7642 _{0.3295}	11.9421 _{1.2181}	5.2062 _{0.3177}	0.2110 _{0.0257}	11.7119 _{0.3891}	0.2270 _{0.0156}	0.0577 _{0.0074}
EDCAG	0.7943 _{0.1775}	4.0004 _{0.3974}	7.3040 _{0.4466}	1.1972 _{0.0610}	1.1715 _{0.0110}	9.2254 _{0.0615}	6.7225 _{0.0161}	6.5375 _{0.3736}	0.4813 _{0.0337}	6.5332 _{0.3092}	0.5281 _{0.0260}	0.2126 _{0.0065}
LFCAG	3.5398 _{1.3093}	72.9698 _{0.3245}	22.5948 _{0.8149}	14.2948 _{0.6703}	15.1854 _{2.4442}	472.0816 _{30.6044}	155.5317 _{2.5999}	709.7923 _{8.2917}	3.4624 _{0.9895}	351.6703 _{1.1724}	4.4383 _{0.3770}	0.2673 _{0.0109}
FCAG	0.6931 _{0.0667}	3.3436 _{0.0444}	4.3694 _{0.0321}	0.9489 _{0.0119}	1.0328 _{0.0751}	9.1004 _{0.5553}	3.8789 _{0.1497}	3.8789 _{0.1497}	0.3758 _{0.0145}	5.9274 _{0.0666}	0.3877 _{0.0147}	0.2305 _{0.0044}
FAGC-R	0.5998 _{0.0410}	4.0143 _{0.3754}	5.3441 _{0.1093}	1.1593 _{0.0681}	1.2247 _{0.0488}	10.1548 _{0.4541}	9.3065 _{0.6034}	4.5893 _{0.0746}	0.2971 _{0.0151}	4.9946 _{0.1681}	0.4780 _{0.0378}	0.2278 _{0.0094}
SC-FAGC(OURS)	0.3371 _{0.0642}	1.4979 _{0.0642}	0.2624 _{0.0059}	0.7338 _{0.0121}	0.5643 _{0.0040}	6.7755 _{0.2024}	3.2644 _{0.1302}	2.6511 _{0.0351}	0.0309 _{0.0029}	0.9480 _{0.0269}	0.1363 _{0.0032}	0.0222 _{0.0009}

Introduce the Lagrange multiplier η for the simplex constraint and multipliers $\mu_j \geq 0$ for non-negativity. The Lagrangian is

$$\mathcal{L}(b, \eta, \mu) = \sum_{j=1}^m d_{ij} b_j + \gamma \sum_{j=1}^m b_j^2 - \eta \left(\sum_{j=1}^m b_j - 1 \right) - \sum_{j=1}^m \mu_j b_j. \quad (21)$$

The KKT conditions are

$$d_{ij} + 2\gamma b_j - \eta - \mu_j = 0, \quad (22)$$

$$\sum_{j=1}^m b_j = 1, \quad b_j \geq 0, \quad \mu_j \geq 0, \quad (23)$$

$$\mu_j b_j = 0, \quad j = 1, \dots, m. \quad (24)$$

For an active anchor with $b_j > 0$, complementary slackness gives $\mu_j = 0$, and hence

$$b_j = \frac{\eta - d_{ij}}{2\gamma}. \quad (25)$$

For inactive anchors, $b_j = 0$ and stationarity implies $\mu_j = d_{ij} - \eta \geq 0$, so only anchors with $d_{ij} < \eta$ can be active.

Sort anchors by increasing distance, $d_{i,(1)} \leq \dots \leq d_{i,(m)}$. Here k is the user-specified neighborhood size, the same k used throughout the anchor-graph construction. Fixing k determines the active set as the k nearest anchors; the threshold η is then pinned at $\eta = d_{i,(k+1)}$, the value at which the $(k+1)$ -th coefficient vanishes while the first k remain positive. Substituting Eq. (25) into the simplex constraint over the active set yields

$$1 = \sum_{h=1}^k \frac{d_{i,(k+1)} - d_{i,(h)}}{2\gamma}, \quad 2\gamma = kd_{i,(k+1)} - \sum_{h=1}^k d_{i,(h)}. \quad (26)$$

Therefore, for the sorted anchor index j ,

$$b_{ij} = \begin{cases} \frac{d_{i,(k+1)} - d_{ij}}{kd_{i,(k+1)} - \sum_{h=1}^k d_{i,(h)}}, & j \leq k, \\ 0, & j > k, \end{cases} \quad (27)$$

The denominator $kd_{i,(k+1)} - \sum_{h=1}^k d_{i,(h)} = \sum_{h=1}^k (d_{i,(k+1)} - d_{i,(h)})$ is strictly positive whenever the $(k+1)$ nearest distances are not all equal, which holds generically; hence $b_{ij} \geq 0$ and the assignment is well defined. This is the closed-form assignment in Eq. (3). The derivation shows how the regularization parameter γ is eliminated by the simplex constraint once the number of active nearest anchors is fixed.

L. Detailed Proofs

L.1. Proof of Lemma 2.1

Proof. Since $f(U) = \text{Tr}(U^\top AU)$ is convex, its first-order Taylor expansion at $U^{(t)}$ provides a global lower bound (subgradient inequality): $f(U) \geq f(U^{(t)}) + \langle \nabla f(U^{(t)}), U - U^{(t)} \rangle$. Computing the gradient $\nabla f(U) = 2AU$, we have:

$$\begin{aligned} f(U) &\geq \text{Tr}(U^{(t)\top} AU^{(t)}) + \text{Tr}(2U^{(t)\top} A(U - U^{(t)})) \\ &= \text{Tr}(2U^{(t)\top} AU) - \text{Tr}(U^{(t)\top} AU^{(t)}). \end{aligned} \quad (28)$$

Adding $\epsilon H(U)$ to both sides yields $\mathcal{J}(U) \geq \mathcal{G}(U \mid U^{(t)})$. The equality at $U = U^{(t)}$ holds trivially. $\square$

L.2. Proof of Theorem 2.2

Proof. By the properties of the MM procedure:

$$\begin{aligned} \mathcal{J}(U^{(t+1)}) &\geq \mathcal{G}(U^{(t+1)} \mid U^{(t)}) \\ &\geq \mathcal{G}(U^{(t)} \mid U^{(t)}) = \mathcal{J}(U^{(t)}), \end{aligned} \quad (29)$$

where the first inequality follows from the minorization property (Lemma 2.1), the second from the maximization step, and the equality from tightness at the anchor point. Since $\mathcal{C}$ is compact and $\mathcal{J}(U)$ is continuous, the objective value is bounded. By the Monotone Convergence Theorem, $\{\mathcal{J}(U^{(t)})\}$ converges. $\square$

L.3. Proof of Theorem 2.3

Proof. A point U^* is a first-order stationary point of (5) if it satisfies the variational inequality

$$\langle \nabla \mathcal{J}(U^*), U - U^* \rangle \leq 0, \quad \forall U \in \mathcal{C}, \quad (30)$$

where $\nabla \mathcal{J}(U) = 2AU + \epsilon \nabla H(U)$. At iteration t , $U^{(t+1)}$ is the unique maximizer of the strictly concave surrogate $\mathcal{G}(\cdot \mid U^{(t)})$ over $\mathcal{C}$ by Proposition 2.4, and therefore satisfies the first-order optimality condition

$$\langle \nabla_U \mathcal{G}(U^{(t+1)} \mid U^{(t)}), U - U^{(t+1)} \rangle \leq 0, \quad \forall U \in \mathcal{C}. \quad (31)$$

Write the subproblem solution as a map of the linearized cost $D = 2AU^{(t)}$. Since the subproblem objective $\langle D, U \rangle + \epsilon H(U)$ is jointly continuous in (D, U) , $\mathcal{C}$ is a nonempty compact set, and the maximizer is unique for every D , Berge's Maximum Theorem (Berge, 1963) implies that the solution map $D \mapsto U^+(D)$ is continuous. As $D = 2AU^{(t)}$ depends continuously on $U^{(t)}$, the composite map $U^{(t)} \mapsto U^{(t+1)}$ is continuous.

Let U^* be any limit point of $\{U^{(t)}\}$ and take a convergent subsequence $U^{(t_\ell)} \rightarrow U^*$. By continuity of the solution map, the successors satisfy $U^{(t_\ell+1)} \rightarrow U^*$. Since

$$\nabla_U \mathcal{G}(U \mid \bar{U}) = 2A\bar{U} + \epsilon \nabla H(U), \quad (32)$$

we have

$$\nabla_U \mathcal{G}(U^{(t_\ell+1)} \mid U^{(t_\ell)}) \rightarrow 2AU^* + \epsilon \nabla H(U^*) = \nabla \mathcal{J}(U^*). \quad (33)$$

Passing to the limit in (31) along this subsequence yields

$$\langle \nabla \mathcal{J}(U^*), U - U^* \rangle \leq 0, \quad \forall U \in \mathcal{C}, \quad (34)$$

which is exactly the first-order KKT condition for (5). $\square$

L.4. Proof of Proposition 2.4

Proof. The objective of the subproblem is linear plus entropic (strictly concave). The sum is strictly concave, guaranteeing a unique maximum. As established in the DB-OT theory (Shi et al., 2024), the dual problem of the subproblem is an unconstrained maximization of a strictly concave semi-dual function. The coordinate ascent on the dual variables corresponds exactly to the scaling updates in Algorithm 1. Since the dual objective is smooth and strictly concave, the block coordinate ascent (Sinkhorn iterations) is guaranteed to converge to the global optimum of the subproblem. $\square$